FACTORIZED CONFORMAL PREDICTION VIA LEARNED RESIDUAL-LAW RETRIEVAL

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ABSTRACT

Conformal prediction provides finite-sample marginal coverage, but constructing locally reliable and efficient multivariate prediction sets remains challenging in high dimensions. We establish a connection between local geometry and score pivotality by showing that approximating query-dependent oracle geometries enclosing the same conditional probability mass across queries yields approximately pivotal nonconformity scores, linking geometry error to local coverage under global calibration. Approximate pivotality also bounds coverage loss under covariate shift without score reweighting. Motivated by this principle, we introduce Factorized Conformal Prediction (FacCP), which separates query-dependent residual geometry from global calibration of the induced scalar score. We realize this factorization through energy-score-trained residual-law retrieval, assuming conditional residual laws admit a low-dimensional uncertainty-sufficient representation. The resulting query-specific residual law determines local geometry and supports multiple set families without retraining, while a single global conformal quantile ensures finite-sample marginal coverage. Controlled experiments validate the geometry–pivotality connection and show that local quantile calibration cannot recover misspecified geometry. In high-dimensional reachabile set prediction, FacCP yields substantially tighter sets, particularly against direct geometry models in low-data and low-capacity regimes, while reducing prediction tube footprints for trajectory containment in autonomous driving.

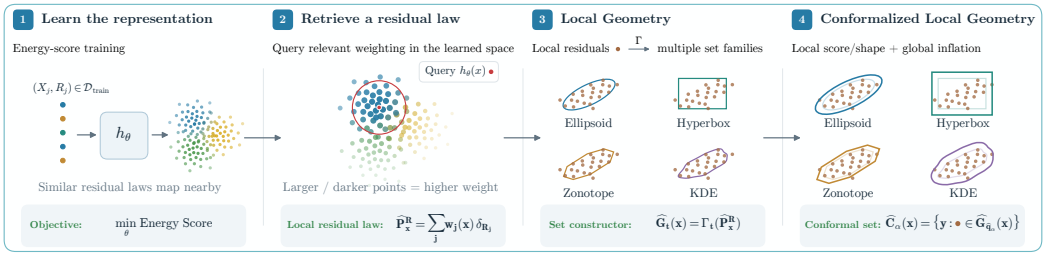


Figure 1: Overview of FacCP.

1 INTRODUCTION

Uncertainty quantification (UQ) constructs high-probability prediction sets for reliable downstream decision-making. Even with limited data, these sets should achieve reliable coverage while remaining tight, globally valid, and locally tailored to each query. Overly conservative sets can eliminate feasible maneuvers in autonomous driving or make robust control inefficient. This adaptivity is difficult when uncertainty varies across operating conditions and local data is scarce, especially in high-dimensional input spaces where naïve localization to nearby examples can further reduce effective sample size. We thus seek prediction sets that adapt to query-dependent uncertainty while making effective use of limited data.

Conformal prediction (CP) converts a fixed nonconformity score into finite-sample marginal coverage under exchangeability, but marginal validity need not imply locally appropriate or geometrically efficient sets. When residual laws vary in orientation, dependence, or modality, a fixed residual geometry may be overly conservative. Learned and normalized scores adapt scalar uncertainty or rectify its conditional distribution (Romano et al., 2019; Seedat et al., 2023; Laplante, 2026; Plassier et al., 2025a), but retain the geometric structure of their base scores. Localized CP adjusts calibration or scores using query-dependent information (Guan, 2023; Hore & Barber, 2025; Kiyani et al.,

2024), although local tail estimation may have limited effective support. Multivariate approaches learn covariances, transformations, or densities (Messoudi et al., 2022; Fang et al., 2025; English & Lippert, 2026), but typically learn a specific uncertainty model or score geometry. These methods address different aspects of adaptivity, but leave open a direct characterization of when approximating query-dependent residual geometry makes scores comparable across queries, allowing a single global calibration threshold to have approximately the same conditional probability content locally.

To address this gap, we establish that approximating query-dependent oracle residual geometries with invariant conditional probability content yields approximately pivotal nonconformity scores. This connects local geometry error to conditional coverage under a single global conformal threshold and bounds coverage loss under covariate shift. Motivated by this principle, we introduce Factorized Conformal Prediction (FacCP). FacCP factorizes conformal set construction into query-dependent residual geometry, which induces a scalar nonconformity score, and a globally calibrated inflation level. We realize this factorization by learning an uncertainty-relevant retrieval representation using the strictly proper energy score of weighted empirical residual distributions. The retrieved law supports multiple query-dependent set geometries without retraining, while one quantile from the full calibration sample determines their inflation and preserves finite-sample marginal validity. We characterize uncertainty-sufficient representations and show how removing irrelevant variation can improve residual-law estimation efficiency. Experiments validate the geometry-to-pivotality connection and demonstrate efficient prediction sets in high-dimensional reachabile set prediction, particularly with limited data or model capacity. Our specific contributions are:

1. We establish a connection between local geometry and score pivotality, showing that approximating oracle query-dependent geometries with invariant conditional probability content bounds score non-pivotality, local coverage error under global calibration, and coverage loss under covariate shift.
2. 2. We introduce Factorized Conformal Prediction (FacCP), separating query-dependent residual geometry from global calibration of the induced scalar score. Assuming a low-dimensional uncertainty-sufficient representation, we instantiate this factorization through energy-score trained residual-law retrieval, characterize the benefits of suppressing irrelevant variation, and support multiple set families without retraining.
3. We validate the geometry–pivotality connection and show that local quantile calibration cannot recover misspecified geometry. FacCP yields tighter calibrated sets in high-dimensional reachabile set prediction, particularly in low-data and low-capacity regimes, and reduces prediction-tube footprints in autonomous driving at comparable coverage.

2 Related Work

CP guarantees marginal coverage under exchangeability (Vovk et al., 2005; Shafer & Vovk, 2008; Lei et al., 2018), whereas distribution-free pointwise conditional coverage is generally unattainable without assumptions (Lei & Wasserman, 2014; Barber et al., 2021). One response is to make score distributions comparable across queries. Oracle conditional-mass scores provide a precedent for pivotality (Romano et al., 2020). Normalization, quantile-based scores, and predictor adjustment target local difficulty or score homogeneity (Papadopoulos et al., 2008; Romano et al., 2019; Seedat et al., 2023; Colombo, 2024; Gao et al., 2024). More directly, PIT-CP and rectified CP transform existing scores to improve conditional invariance (Laplante, 2026; Plassier et al., 2025a). These transformations retain the candidate ordering defined by a base score within each query. PIT-CP also studies geometric differences between transformed sets, but bounds them via score-distribution estimation error. Our analysis instead bounds the full conditional score CDF discrepancy from approximation of oracle geometric sublevel sets, before specifying how those geometries are learned.

Localized CP adapts calibration weights, partitions, or scores using conditional information (Guan, 2023; Hore & Barber, 2025; Han et al., 2022; Kiyani et al., 2024; Gibbs et al., 2025). This can improve local coverage, but estimating local tail thresholds can reduce effective calibration support. FacCP localizes residual-law estimation rather than the final quantile, then calibrates one score threshold using all calibration observations. Conditional-distribution methods obtain adaptive sets from density or CDF estimates and conditional samples (Lei & Wasserman, 2014; Chernozhukov et al., 2021; Izbicki et al., 2020; 2022; Diamant et al., 2024; Wang et al., 2023). CP2 also supports flexible query-dependent geometries and global calibration, deriving conditional-coverage bounds from conditional-law estimation error (Plassier et al., 2025b). Our theory instead bounds score non-pivotality by probability-mass disagreement between learned and oracle geometric sublevel

sets, without requiring control of the full conditional law in total variation. FacCP realizes this perspective through energy-score-trained residual retrieval in an uncertainty-sufficient query representation, rather than direct conditional density or generator estimation.

Multivariate CP learns ellipsoids, response transformations, templates, transport ranks, or minimum-volume regions (Messoudi et al., 2022; Xu et al., 2024; Feldman et al., 2023; Tumu et al., 2024; Thurin et al., 2025; Braun et al., 2026; Conrad et al., 2026; Kondratyev et al., 2026; Klein et al., 2026). Flow methods such as CONTRA and JAPAN provide flexible score geometries (Fang et al., 2025; English & Lippert, 2026). Unlike direct geometric models, FacCP learns a conditional residual-law retrieval metric, then constructs geometry for different set families without retraining the representation. Weighted CP addresses covariate shift through calibration weights (Tibshirani et al., 2019; Barber et al., 2023; Joshi et al., 2026); approximate pivotality instead supports reusing an unweighted threshold across shifts preserving the conditional response law, subject to the geometry error. App. B discusses these distinctions further.

3 Problem Statement

Let $(X, Y) \sim P_{XY}$, where $X \in \mathcal{X}$ is a query and $Y \in \mathcal{Y}$ is its response. Given a fixed predictor, let $T\colon \mathcal{X} \times \mathcal{Y} \to \mathcal{R}$ be a measurable, task-specific residual map, where $\mathcal{R} \subseteq \mathbb{R}^{d_R}$. Define

$$R := T(X, Y), \qquad P_x^R := \mathcal{L}(R \mid X = x). \tag{1}$$

The conditional residual law P_x^R characterizes uncertainty at x, allowing location, scale, dependence, and shape to vary across queries. The residual map T is fixed, whereas the scalar nonconformity score is induced by a query-dependent residual geometry (Sec. 4.1).

We are given disjoint reference and calibration samples $\mathcal{D}_{\mathrm{ref}} = \{(X_j^{\mathrm{ref}}, Y_j^{\mathrm{ref}})\}_{j=1}^m$ and $\mathcal{D}_{\mathrm{cal}} = \{(X_i^{\mathrm{cal}}, Y_i^{\mathrm{cal}})\}_{i=1}^n$. The reference data are used to construct nonconformity scores, which are fixed before calibration. The calibration observations and a future test observation $(X_{\mathrm{new}}, Y_{\mathrm{new}})$ are assumed exchangeable conditional on all data and randomness used to construct the scores. For a target miscoverage level $\alpha \in (0, 1)$, we seek prediction sets $\widehat{C}_\alpha(x) \subseteq \mathcal{Y}$ satisfying finite-sample marginal coverage

$$\mathbb{P}\big(Y_{\mathrm{new}} \in \widehat{C}_\alpha(X_{\mathrm{new}})\big) \geq 1 - \alpha. \tag{2}$$

Marginal validity follows from split-conformal calibration of any fixed nonconformity score, but does not guarantee reliable conditional coverage. We first characterize when query-dependent residual geometries induce approximately pivotal scores, so a single pooled calibration quantile has comparable conditional probability content across queries. We then quantify how geometry approximation affects local coverage and coverage under covariate shift. Realizing this principle requires estimating local geometries efficiently. To this end, we estimate the conditional residual law P_x^R and construct query-dependent geometries from it, allowing the same learned representation to support different set families without retraining.

4 Method: FacCP

We overview FacCP (see Fig. 1), which separates query-dependent residual geometry from global calibration of the scalar score induced by that geometry. The central principle is that approximating oracle local geometries with query-invariant conditional probability content yields approximately pivotal scores, allowing a globally calibrated threshold to remain locally meaningful. We first establish this general connection (Sec. 4.1), then develop energy-score-trained residual-law retrieval (Sec. 4.2), and finally construct and calibrate prediction sets from those laws (Sec. 4.3).

4.1 Geometry, Pivotality, and Local Coverage

Conformal prediction guarantees finite-sample marginal coverage for a fixed nonconformity score under exchangeability, but this need not imply locally reliable or geometrically efficient prediction sets. When conditional residual laws vary across queries, a fixed residual geometry may assign different probability mass to the same score threshold. Our starting point is that adapting the geometry can make the resulting scores comparable across queries, allowing local uncertainty to be represented before applying a single global calibration threshold.

Consider a measurable, query-dependent nested family of residual sets $\{\widehat{G}_a(x)\}_{a \geq 0}$, where $\widehat{G}_b(x) \subseteq \widehat{G}_a(x)$ for $0 \leq b \leq a$. The geometry is constructed using auxiliary data independent

of the conformal calibration sample. Each family induces a scalar nonconformity score, defined for a candidate response y as the smallest inflation level containing its residual,

$$\widehat{S}(x, y) := \inf\{a \geq 0 : T(x, y) \in \widehat{G}_a(x)\}. \tag{3}$$

For the constructors considered here, the infimum is attained and the nested families are right-continuous, ensuring that $\widehat{S}(x, y) \leq a$ is equivalent to $T(x, y) \in \widehat{G}_a(x)$. Thus, geometrically different residual sets can be represented by scalar scores with a common inflation parameter. The question is when this parameter has approximately the same probabilistic meaning across queries.

Throughout, we condition on $\mathcal{D}_0$, containing all data and randomness used to construct $\widehat{S}$, and suppress this conditioning for notational simplicity, so $\widehat{S}$ is fixed. Define the conditional and marginal score distributions by $F_x(a) := \mathbb{P}(\widehat{S}(X, Y) \leq a \mid X = x)$ and $F(a) := \mathbb{P}(\widehat{S}(X, Y) \leq a)$. By the correspondence between scores and nested sets, $F_x(a) = P_x^R(\widehat{G}_a(x))$. Let $\mathcal{X}_0 \subseteq \mathcal{X}$ denote the covariate region over which local behavior is assessed. We take $\mathcal{X}_0$ to contain the support of the marginal distribution defining F, as well as any target-relevant covariates considered under shift.

Definition 1 (Pivotality)**.** The score $\widehat{S}(X, Y)$ is pivotal with respect to X on $\mathcal{X}_0$ if $F_x(a) = F(a)$ for every $x \in \mathcal{X}_0$ and $a \geq 0$. We quantify departures from pivotality by $\Delta_{\text{piv}} := \sup_{x \in \mathcal{X}_0,\, a \geq 0} |F_x(a) - F(a)|$.

Pivotality requires every inflation level to contain the same conditional residual probability mass. When $\Delta_{\text{piv}} = 0$, a common threshold therefore has identical conditional probability content. More generally, small Δ_{piv} allows globally pooled calibration scores to remain comparable despite query-dependent geometry.

To characterize when this occurs, consider an oracle nested geometry $\{G_a^\star(x)\}_{a \geq 0}$ with query-invariant conditional mass $P_x^R(G_a^\star(x)) = F_0(a)$ for every $x \in \mathcal{X}_0$ and $a \geq 0$. Such an oracle can be constructed by mapping each P_x^R to a common reference law, when the requisite coordinate maps exist, and applying a common nested family in those coordinates. The formal construction and oracle-pivotality result are given in Thm. 6. FacCP does not estimate these coordinate maps explicitly. Instead, it constructs query-dependent geometries from retrieved conditional residual laws. Perfect recovery of the oracle geometry would induce pivotal scores, but exact geometric recovery is unnecessary when discrepancies occur in regions with negligible conditional probability mass.

Assumption 2 (Uniform geometry approximation)**.** *For some $\varepsilon_{\text{geom}} \geq 0$, and $\triangle$ denoting symmetric difference,* $\sup_{x \in \mathcal{X}_0,\, a \geq 0} P_x^R\Big(\widehat{G}_a(x) \triangle G_a^\star(x)\Big) \leq \varepsilon_{\text{geom}}$.

The quantity $\varepsilon_{\text{geom}}$ measures the worst-case conditional probability mass on which the learned and oracle geometries disagree. Since the score CDF at level a equals the probability mass of the corresponding geometric sublevel set, this approximation directly controls the discrepancy between their score laws.

Theorem 3 (Geometry approximation controls pivotality)**.** *Under Assumption 2,* $\sup_{x \in \mathcal{X}_0,\, a \geq 0} |F_x(a) - F_0(a)| \leq \varepsilon_{\text{geom}}$*, and* $\Delta_{\text{piv}} \leq 2\varepsilon_{\text{geom}}$.

Thm. 3 establishes a sufficient condition for approximate pivotality in terms of geometric approximation. In particular, approximating the oracle sets where the conditional residual law places mass makes the induced score distributions comparable across queries. This result applies to any nested local geometry satisfying Assumption 2, independently of how that geometry is learned. This motivates incorporating query-dependent residual structure into the geometry before scoring, so a single global threshold suffices for calibration.Our implementation learns an uncertainty-relevant retrieval space and constructs local geometries from observed residuals rather than directly predicting high-dimensional geometry parameters, enabling data and parameter efficient construction.

The resulting score comparability also controls local coverage. Define the marginal population quantile $q_\alpha := F^{-1}(1 - \alpha) := \inf\{a : F(a) \geq 1 - \alpha\}$, where F^{-1} is the generalized inverse, and suppose $F(q_\alpha) = 1 - \alpha$. Then

$$\sup_{x \in \mathcal{X}_0} \Big|\mathbb{P}\Big(\widehat{S}(X, Y) \leq q_\alpha \mid X = x\Big) - (1 - \alpha)\Big| = \sup_{x \in \mathcal{X}_0} \Big|P_x^R\Big(\widehat{G}_{q_\alpha}(x)\Big) - (1 - \alpha)\Big| \leq \Delta_{\text{piv}}. \tag{4}$$

See Cor. 7 for a formal proof. Thus, Δ_{piv} bounds the structural worst-case local coverage error even with the exact marginal population quantile. Estimating this quantile from a finite calibration

sample introduces an additional error of order $n^{-1/2}$ under the conditions of Thm. 8. Combined with Thm. 3, this separates local coverage error into geometry induced non-pivotality and global calibration error. Since Δ_{piv} is a population supremum rather than a generally estimable uniform certificate in continuous settings, we use it primarily as an empirical diagnostic.

Transfer Under Covariate Shift. Approximate pivotality naturally supports covariate-shift transfer. Under shifts preserving $P_{Y|X}$, with source and target covariates in $\mathcal{X}_0$, the source-calibrated conformal inflation satisfies $P^{\mathrm{target}}(Y_{\mathrm{new}} \in \widehat{C}_\alpha(X_{\mathrm{new}})) \geq 1 - \alpha - \Delta_{\mathrm{piv}}$, where Δ_{piv} is defined under the source distribution. Under Assumption 2, the coverage loss is at most $2\varepsilon_{\mathrm{geom}}$. See Cor. 9.

The remaining task is to construct local geometries that approximate the conditional residual law without requiring separate calibration thresholds at every query. FacCP addresses this by learning which observed residuals are informative about each query's conditional residual law.

4.2 Learning the Conditional Residual Law

Let P_x^R denote the conditional residual law in (1). When P_x^R varies with x, a fixed residual geometry must accommodate errors plausible under different local laws, potentially producing conservative sets. FacCP estimates P_x^R as $\widehat{P}_x^R$ and subsequently maps this estimate into a query-dependent geometry $\widehat{G}_a(x)$. Separating residual-law estimation from geometry construction allows one estimated distribution to support different set families without retraining the uncertainty representation.

We estimate P_x^R by reweighting observed residuals from $\mathcal{D}_{\mathrm{ref}}$, where $R_j = T(X_j^{\mathrm{ref}}, Y_j^{\mathrm{ref}})$. Direct input-space distances can be misleading because distant inputs may induce similar residual laws, while nearby inputs may have different uncertainty. We therefore learn a representation $h_\theta : \mathcal{X} \to \mathbb{R}^p$, writing $Z = h_\theta(X)$ and $z = h_\theta(x)$, in which proximity is intended to reflect similarity of conditional residual distributions. Given a query x, define normalized retrieval weights

$$w_j(x) = \frac{\exp\{-d_Z(h_\theta(x), h_\theta(X_j))/\tau\}}{\sum_{k=1}^m \exp\{-d_Z(h_\theta(x), h_\theta(X_k))/\tau\}}, \qquad \sum_{j=1}^m w_j(x) = 1, \tag{5}$$

where d_Z is a metric in latent space and $\tau > 0$ controls the locality of weighting. The resulting conditional residual-law estimator is the weighted empirical distribution

$$\widehat{P}_x^R = \sum_{j=1}^m w_j(x)\delta_{R_j}. \tag{6}$$

$\widehat{P}_x^R$ remains a distribution on the original residual space $\mathcal{R}$, and h_θ determines which reference residuals contribute to its estimation. Thus, the learned coordinates solve a localization problem rather than directly predicting high-dimensional geometry . Its purpose is to preserve information about how the conditional residual law changes with the query.

For residual modeling, Z need not retain every feature of X needed to predict Y. It must retain the information determining the conditional law of R. We call Z uncertainty-sufficient if $R \perp\!\!\!\perp X \mid Z$, or equivalently, since $Z = h_\theta(X)$ is deterministic, if $\mathcal{L}(R \mid X) = \mathcal{L}(R \mid Z)$ almost surely. Conditioned on an uncertainty-sufficient representation, the remaining information in X does not change the residual distribution. This permits queries with different input features but identical residual laws to share reference observations.

To characterize the information retained by the representation, we use the energy score. For a law Q on $\mathcal{R} \subseteq \mathbb{R}^{d_R}$ with finite first moment and an observation $r \in \mathcal{R}$, define $\mathrm{ES}(Q, r) = \mathbb{E}_{V\sim Q}\|V - r\|_2 - \frac{1}{2}\mathbb{E}_{V,V'\overset{\text{i.i.d.}}{\sim}Q}\|V - V'\|_2$. The first term measures agreement with the realized residual, while the second accounts for the dispersion of the predicted law. The energy score is strictly proper (Gneiting & Raftery, 2007, Thm. 5), so its expected value is uniquely minimized by the data-generating distribution, $P = \arg\min_Q \mathbb{E}_{R\sim P}[\mathrm{ES}(Q, R)]$. Consequently, its unrestricted conditional population target is the full law P_x^R.

This property also characterizes uncertainty sufficiency. Define $P_X^R := \mathcal{L}(R \mid X)$ and $P_Z^R := \mathcal{L}(R \mid Z)$. The former conditions on the complete query, while the latter is the residual law available after representation compression. For probability laws P, Q with finite first moments, the energy score satisfies $\mathbb{E}_{R\sim P}[\mathrm{ES}(Q, R)] - \mathbb{E}_{R\sim P}[\mathrm{ES}(P, R)] = \frac{1}{2}\mathcal{E}(P, Q)$, where $\mathcal{E}(P, Q) \geq 0$ is the energy distance, which vanishes if and only if $P = Q$. Applying this identity conditionally yields the following representation-specific result.

Proposition 4 (Energy-score cost)**.** *Assume the conditional residual laws have finite first moments, and define* $\mathcal{L}_X^\star = \mathbb{E}[\mathrm{ES}(P_X^R, R)]$ *and* $\mathcal{L}_Z^\star = \mathbb{E}[\mathrm{ES}(P_Z^R, R)]$*. Then* $\mathcal{L}_Z^\star - \mathcal{L}_X^\star = \frac{1}{2}\mathbb{E}\big[\mathcal{E}(P_X^R, P_Z^R)\big]$*. Thus* $\mathcal{L}_Z^\star = \mathcal{L}_X^\star$ *if and only if* $P_X^R = P_Z^R$ *almost surely, equivalently, if and only if* $R \perp\!\!\!\perp X \mid Z$*.*

Prop. 4 establishes that compression is lossless for residual-law modeling exactly when the representation is uncertainty-sufficient. Otherwise, its excess optimal energy-score risk equals $\frac{1}{2}\mathbb{E}[\mathcal{E}(P_X^R, P_Z^R)]$. The representation should therefore preserve variations in X that change the residual law while suppressing irrelevant variation. This gives a population characterization of what the retrieval coordinates should retain.

We now use the energy score to learn these coordinates. Input-space proximity alone cannot determine whether queries have similar residual laws, so the representation is trained according to the distributional quality of its retrieved residuals. For a training sample (x_i, r_i), let J_i denote the reference indices excluding the query itself. Define the training retrieval weights and corresponding empirical residual law by

$$w_{ij}^{(\theta)} = \frac{\exp\{-d_Z(h_\theta(x_i),h_\theta(X_j))/\tau\}}{\sum_{k\in J_i}\exp\{-d_Z(h_\theta(x_i),h_\theta(X_k))/\tau\}}, \quad j \in J_i, \qquad \widehat{P}^R_{x_i,\theta} = \textstyle\sum_{j\in J_i} w_{ij}^{(\theta)}\delta_{R_j}. \tag{7}$$

Excluding observation i prevents trivial self-retrieval of r_i. We train h_θ by minimizing

$$\ell_i(\theta) = \textstyle\sum_{j\in J_i} w_{ij}^{(\theta)}\|R_j - r_i\|_2 - \frac{1}{2}\sum_{j,k\in J_i} w_{ij}^{(\theta)} w_{ik}^{(\theta)}\|R_j - R_k\|_2. \tag{8}$$

By direct substitution into the energy-score definition, $\ell_i(\theta) = \mathrm{ES}(\widehat{P}^R_{x_i,\theta}, r_i)$, as formalized in Prop. 11. The first term favors residuals compatible with r_i, but alone can collapse retrieval toward one residual. The second accounts for dispersion, preventing this degeneracy; together, they score the weighted residuals as a probability distribution. Each r_i is a sample from the unknown conditional law $P^R_{x_i}$, which the retrieved distribution is trained to predict.

By strict propriety, the unrestricted population objective is minimized by the true conditional residual law almost surely under P_X. Thus, the objective favors accurate conditional residual distribution retrieval, rather than directly modeling high dimensional geometry. Together with Prop. 4, this motivates learning a representation that preserves information relevant to the conditional residual law.

Uncertainty sufficiency also has consequences for estimation efficiency. Removing directions that affect retrieval distances but not the residual law can reduce the effective dimension of localization. Under the stylized local-kernel model with regular local geometry and Hölder-smooth conditional residual laws in Prop. 12, suppose an uncertainty-sufficient localization variable has intrinsic dimension d, while $p-d$ additional directions vary under the retrieval metric but are conditionally irrelevant to R. Writing $\mathfrak{R}_s^\star$ for the optimally tuned worst-case expected energy-distance error when localizing in intrinsic dimension s, the proposition gives $\mathfrak{R}_d^\star \asymp m^{-2\beta/(2\beta+d)}$ and $\mathfrak{R}_p^\star \asymp m^{-2\beta/(2\beta+p)}$ for $p > d$. Thus, uncertainty-irrelevant variation can slow residual-law estimation without contributing information about P_x^R, as illustrated empirically in Fig. 2. Learning a compact representation can improve localization when it suppresses these directions, while excessive compression may discard residual-law information and incur the representation error characterized in Prop. 4.

4.3 From Residual Distributions to Conformal Sets

The retrieved law $\widehat{P}_x^R$ provides a query-dependent estimate of residual uncertainty but is not itself a prediction set. To construct the geometries analyzed in Sec. 4.1, let Γ be a geometry constructor mapping any probability law Q on $\mathcal{R}$ to a nested family $\{\Gamma_a(Q)\}_{a\geq 0}$ satisfying $\Gamma_b(Q) \subseteq \Gamma_a(Q)$ for $0 \leq b \leq a$. We define $\widehat{G}_a(x) := \Gamma_a(\widehat{P}_x^R)$. The retrieved law determines the geometry at each query, while the index a controls its inflation. Nestedness ensures that these sets induce the nonconformity score $\widehat{S}$ in (3). The constructor can be selected according to the residual space and downstream task without changing the learned retrieval representation. Even exact recovery of P_x^R does not ensure Assumption 2, since a restricted geometry constructor may fail to approximate oracle sublevel sets with query-invariant conditional probability mass.

For example, an ellipsoidal constructor summarizes $\widehat{P}_x^R = \sum_{j=1}^m w_j(x)\delta_{R_j}$ through its weighted center $\widehat{c}(x) = \sum_{j=1}^m w_j(x)R_j$ and regularized weighted covariance $\widehat{\Sigma}(x) = \sum_{j=1}^m w_j(x)\big(R_j - \widehat{c}(x)\big)\big(R_j - \widehat{c}(x)\big)^\top + \lambda I$, for $\lambda > 0$. The resulting nested family is $\widehat{G}_a(x) = \{r \in \mathcal{R} : (r - \widehat{c}(x))^\top\widehat{\Sigma}(x)^{-1}(r - \widehat{c}(x)) \leq a^2\}$. Thus, $\widehat{P}_x^R$ determines the local center, orientation, and relative scale, while a controls overall inflation. Other constructors, including hyperboxes, zonotopes, and non-convex density-based sets, can use the same retrieved law without retraining h_θ. This separation works because the representation retrieves residual distributions rather than predicting parameters for one set family.

We then calibrate the induced scalar score globally. Freeze $\widehat{S}$ using all data and randomness independent of the calibration sample $\mathcal{D}_{\text{cal}} = \{(X_i, Y_i)\}_{i=1}^n$. Since query dependence is already encoded in the geometry, no local quantile estimation is required. Compute $S_i = \widehat{S}(X_i, Y_i)$ for each calibration observation, let $S_{(1)} \le \cdots \le S_{(n)}$ denote the ordered scores, and define $S_{(n+1)} := \infty$. For target miscoverage $\alpha \in (0, 1)$, set $k = \lceil (n+1)(1-\alpha) \rceil$ and $\widehat{q}_\alpha := S_{(k)}$. The resulting prediction set is $\widehat{C}_\alpha(x) = \{y \in \mathcal{Y} : \widehat{S}(x, y) \le \widehat{q}_\alpha\} = \{y \in \mathcal{Y} : T(x, y) \in \widehat{G}_{\widehat{q}_\alpha}(x)\}$.

All n calibration observations contribute to one global inflation quantile despite differing local geometries. Reference data determine query-specific uncertainty geometry, while independent calibration data determine the common inflation level required for coverage. Under exchangeability, standard split-conformal validity guarantees $\mathbb{P}(Y_{\text{new}} \in \widehat{C}_\alpha(X_{\text{new}})) \ge 1 - \alpha$, as formalized in Thm. 10. This marginal guarantee does not depend on the accuracy of $\widehat{P}_x^R$ or $\widehat{G}_a(x)$. Retrieval and geometry approximation instead determine the efficiency of the resulting sets and the comparability of their scores across queries.

The analysis in Sec. 4.1 explains when this globally calibrated construction is also locally reliable. Under Assumption 2, Thm. 3 bounds the geometry-induced score discrepancy by $\Delta_{\text{piv}} \le 2\varepsilon_{\text{geom}}$. The population local-coverage bound in (4) then applies to the common population quantile, while finite calibration adds only the global quantile-estimation error characterized in Thm. 8. Thus, local residual-law retrieval determines the geometry used for scoring, and approximate pivotality determines whether the globally estimated inflation has comparable conditional probability content across queries.

5 Numerical Experiments

5.1 Controlled Residual Geometry and Localization

We first isolate the role of residual geometry and localization in a setting with an analytic conditional residual law. Let $X = (X_1, X_2, X_3, N)$, where the first three coordinates independently control residual scale, aspect ratio, and orientation, and N contains optional nuisance coordinates independent of the residual law. We draw $R \mid X = x \sim \mathcal{N}(0, \Sigma(x))$. The nuisance coordinates affect distances in the observed input space but do not affect $\Sigma(x)$ or the conditional residual law P_x^R.

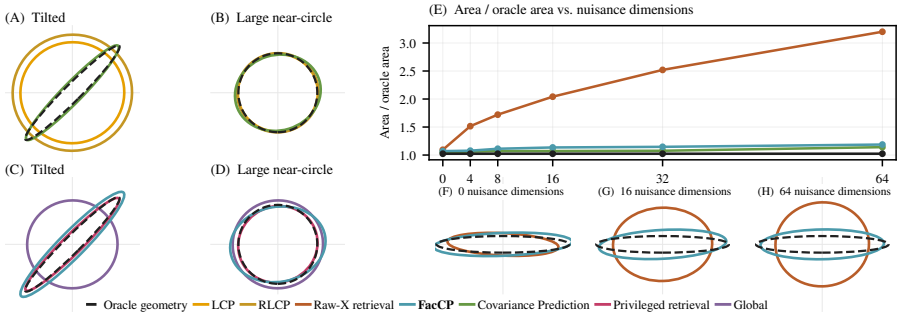


Figure 2: Controlled residual-geometry experiment. (A–D) show prediction sets under multiple methods. (E) shows that FacCP stays close to oracle area compared to raw-X retrieval under nuisance dimensions. (F–H) show the degradation of raw-X retrieval at multiple nuisance dimensions. Fig. 2 compares six geometry constructions. Oracle is the true $\Sigma(x)$, Global uses one covariance for all queries, Privileged retrieval localizes with the known uncertainty-generating coordinates, Raw-X retrieval localizes directly in X, FacCP learns the retrieval representation via the energy-score objective, and covariance prediction directly estimates $\Sigma(x)$ by Gaussian likelihood.

Without nuisance coordinates, FacCP achieves mean area $1.07\times$ oracle, compared with $3.95\times$ for Global. Panels A–B show that Localized-CP methods (LCP (Guan, 2023), RLCP (Hore & Barber, 2025)) reduce heterogeneity in the global score but remain limited to globally oriented ellipses. Even with the pointwise conditional 95% score quantile, mean area is $1.99\times$ oracle, showing that local calibration corrects scalar variation but not query-dependent aspect ratio or orientation.

Panels E–H illustrate that Raw-X retrieval works well without nuisance coordinates but degrades as uncertainty-irrelevant variation grows, consistent with Prop. 12. At 16 nuisance dimensions, its mean area is $2.04\times$ oracle with worst conditional-coverage deviation 0.301, versus $1.14\times$ and 0.066 for FacCP; at 64 dimensions, raw-X reaches $3.20\times$ while FacCP remains at $1.19\times$. Covariance

prediction remains strong in this data-rich, well-specified setting; This improved local geometry also yields a lower score pivotality defect (0.117 vs. 0.361 for Global). Additional pivotality diagnostics and baselines are in App. C.1.

Covariate Shift We test the covariate-shift transfer result of Cor. 9 by calibrating once on the source distribution and evaluating under shifts that preserve $P_{R|X}$. With $U_1, U_2, U_3 \sim \mathrm{Unif}[-1, 1]$ controlling residual scale, aspect ratio, and orientation, we tilt one coordinate at test time via $p_\beta(u_j) \propto \exp(\beta u_j)$, with $\beta = 0$ recovering the source. Under scale and aspect-ratio shifts, Global coverage degrades to 87.3% and 89.2% at $\beta = 3$, while FacCP remains near the nominal 95% target at 94.9% and 96.4%. Under orientation shift, both remain near nominal coverage because the pooled Global geometry is roughly rotation-symmetric here. Full sweeps and baselines are in App. C.2.

5.2 Quadcopter Reachability

2D Quadcopter. We evaluate FacCP for conformal reachability on 100-step planar quadcopter rollouts. Each query comprises a nominal 6-state trajectory and a 2-dimensional control sequence, giving an 806-dimensional input and a 600-dimensional residual trajectory. For ellipsoidal sets, the 100 time-indexed centers and covariance matrices require 2,700 geometric parameters per query. We target simultaneous 99% coverage of the full trajectory. Rather than calibrating each time step separately and applying a Bonferroni correction, we follow the single nonconformity score approach of Lindemann et al. (2025), which aggregates time-indexed geometric scores into one trajectory-level score and conformalizes a single threshold. FacCP constructs query-dependent geometries from retrieved residual laws, while retaining one globally calibrated trajectory-level threshold.

At comparable coverage, these experiments evaluate one representation across geometries, local versus global set efficiency, and FacCP's data and parameter efficiency relative to direct geometry models. See App. C.3 for details.

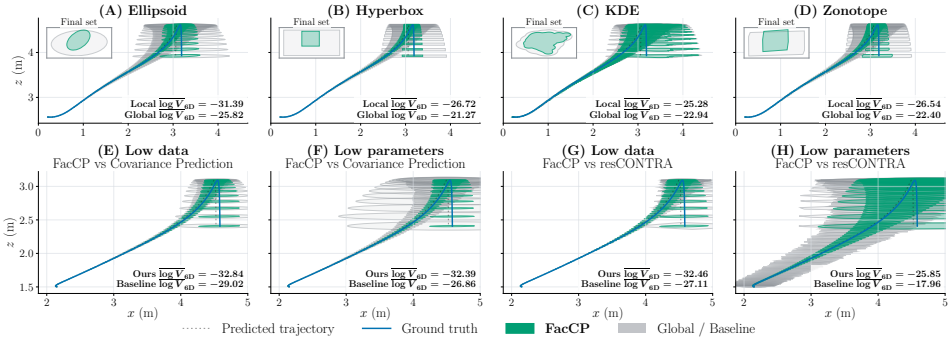


Figure 3: (A-D) One learned representation supports multiple geometries, locally efficient vs global reachable sets; (E-H) FacCP outperforms baselines in low data and model parameter regimes.

Fig. 3 (A-D) shows that one h_θ supports ellipsoidal, hyperbox, KDE, and zonotopic sets without retraining. Across constructors, local geometry reduces log volume by 2.34 to 5.57 relative to global geometry. For ellipsoids, retrieval in the raw and standardized 806D input gives log volumes -26.66, close to the global -25.82, whereas learned retrieval reaches -33.25 with 99.1% coverage. Additional ablation further shows that energy-score training reduces mean log-volume by 5.64–7.43 relative to covariate-reconstruction and residual-prediction autoencoders across 90–99% coverage targets (Table 6). This matches Prop. 4 and Prop. 12. An uncertainty-sufficient representation discards variation that does not change the residual law. Removing such variation increases effective local support and lowers the intrinsic localization dimension. Thus FacCP solves the same residual-law estimation problem in a more favorable localization space. See App. C.3 for further details.

We compare FacCP with resCONTRA (Fang et al., 2025) and Covariance Prediction CP (CVCP) (Srinivasan et al., 2026; Messoudi et al., 2022) across parameter counts, matching the same SNSA calibration procedure; full results appear in Table 7. resCONTRA learns a query-dependent flow, while CVCP directly predicts covariance. At roughly 5×10^3 parameters, FacCP yields sets over $10^3\times$ smaller than resCONTRA at comparable coverage, and at low capacity over $250\times$ smaller than CVCP (Fig. 3 (F & H). These gaps close with substantially greater model capacity. Prop. 11 establishes that the retrieval objective targets the conditional residual law, encouraging distributionally relevant pooling rather than potentially parameter-intensive direct geometry prediction.

We next fix parameter count near each previous sweep's crossover and vary the training-data budget; full results are in Table 8. At 20% of the data, FacCP produces sets $211\times$ smaller than resCONTRA

and $46\times$ smaller than CVCP while maintaining coverage (Figure 3 (E & G)); these gaps shrink with more data. Consistent with Props. 11, 4, and 12, FacCP learns an uncertainty-relevant retrieval metric and constructs geometry from retrieved residuals, whereas direct methods parameterize the query-dependent geometry itself. This favors FacCP in low-data, low-capacity regimes, although with sufficiently scaled data and model capacity, direct estimators can outperform it.

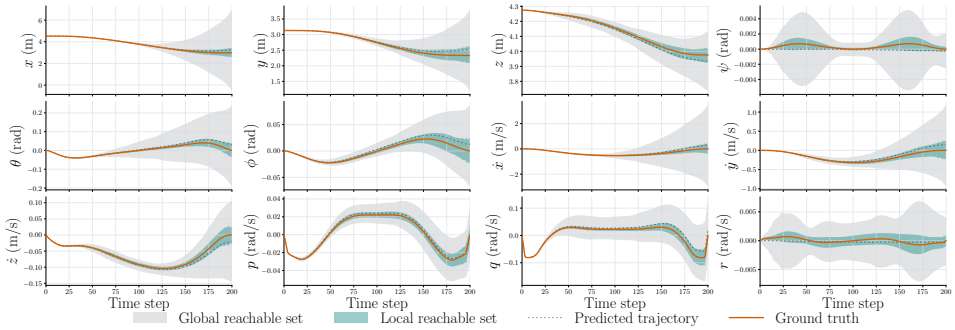


Figure 4: 3D quadcopter conformal reachability for 200 timesteps.

3D Quadcopter. We scale the experiment to 200-step rollouts of a 12-state, 4-control 3D quadcopter, yielding 3212-dimensional queries and 2400-dimensional residual trajectories. The 200 time-indexed ellipsoids require 18,000 geometric parameters per query, comprising 2,400 mean and 15,600 independent covariance parameters. Geometry construction and trajectory-level calibration follow the planar setup, targeting 99% simultaneous full-trajectory coverage with one conformal event. Figure 4 shows local retrieval remains effective at this scale, reducing mean per-time-step 12D log volume from -52.52 to -68.99 at comparable coverage (99.4% global vs. 99.3% local).

The model capacity and data sweeps sharpen this scaling result. FacCP reaches log volumes near -78 with tens of thousands of parameters; Mahalanobis CP remains less efficient even at $\sim 1.6 \times 10^6$ parameters and across every tested data budget. resCONTRA closes the capacity gap between 6×10^4 and 9×10^4 parameters and, at fixed $\sim 6 \times 10^4$ capacity, closes the data gap near 70% of the training budget. Thus retrieval is more efficient in the low-capacity and low-data regimes, while an expressive and sufficiently trained flow can eventually surpass it. Full sweeps are in App. C.3.1.

5.3 Autonomous Driving

Motion forecasting predicts where an agent will go using its past motion, the surrounding agents, and the road map. Its errors are multimodal and depend strongly on the scene, so a planner needs uncertainty sets that adapt to each situation while still covering the true future. We evaluate FacCP on Argoverse 2 (Wilson et al., 2021) around an MTR forecaster trained with annealed Winner-Takes-All (Shi et al., 2022; Xu et al., 2025). Each query is a scene, comprising the focal agent's history, up to 32 neighboring agents, and up to 128 road-map polylines, roughly 3.5×10^4 input coordinates. The aWTA MTR forecaster predicts six candidate trajectories over six seconds, each with a probability. FacCP constructs uncertainty tubes around these fixed paths. Each tube spans 60 timesteps, so every candidate requires a sequence of 60 covariances and error scales.

Method	Mean footprint (m^2·s)	Whole-path coverage (%)
MTR raw-head	1,056.0	4.15
Global SNSA	679.6	95.21
Local SNSA (FacCP)	586.4	94.92
FORCE-OPT	197.6	62.29
MoE-CP (adapt.)	764.1	95.39

Table 1: Six-mode results at the 95% target

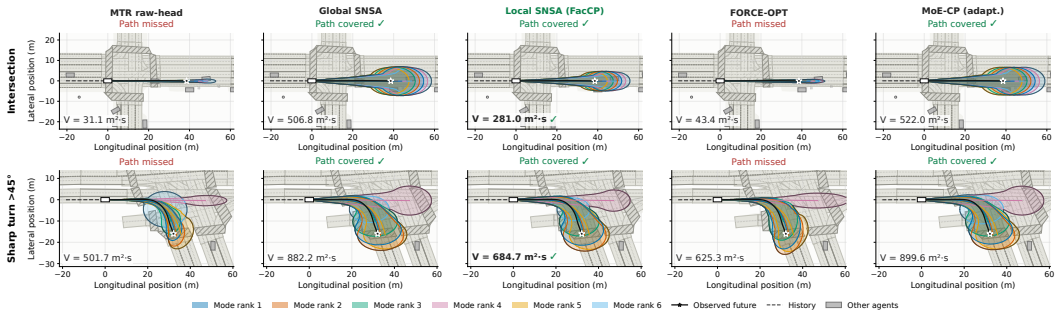


Figure 5: Intersection (top) and sharp-turn (bottom) scenes at the 95% target. Local SNSA (FacCP) produces the tightest tubes that cover the complete observed trajectory in each example.

We target whole-path coverage where one of the six tubes must contain the complete future at every timestep, without switching between modes. Rather than calibrating each timestep and applying a Bonferroni correction, we follow the quadcopter setup and use a single SNSA score, the maximum normalized error over time minimized over candidates, with one conformal threshold (App. D.3). Global SNSA estimates each timestep's covariance and 95th-percentile Mahalanobis error scale from all reference scenes with equal weight, and shares them across scenes and candidates. Our Local SNSA (FacCP) estimates the same quantities from reference errors weighted by learned embedding similarity, so each candidate tube in each scene receives its own shape and width. The two differ only in how reference errors are weighted. Both methods add a pricing fee $\propto -\log \pi_k$, shifting tube budget toward probable candidates; App. D.12 isolates its effect. FacCP's encoder is trained in two stages: masked autoencoder pretraining followed by energy-score fine-tuning. Implementation details are in App. D.2 and D.6. To assess the richness of our FacCP embeddings, we run the SNSA construction with FacCP embeddings without energy-score finetuning & MTR features. We also replace energy fine-tuning with a KL objective to isolate the effect of the training objective. Our energy-trained FacCP yields smaller footprints at same nominal coverage targets in both comparisons (App D.4).

Next we compare FacCP with FORCE-OPT (Chakraborty et al., 2026) and MoE-CP (Kong et al., 2026), both adapted to the same MTR predictions. FORCE-OPT calibrates reachable sets for safe-planning evaluation at each timestep, which does not imply that one tube covers the whole path. Guaranteeing 95% over all 60 steps by Bonferroni requires roughly 99.9% per step. MoE-CP localizes the calibration threshold by training a mixture-of-experts and passing them through a gating network to produce an embedding for each scene used for calculating reference weights. We use MTR's six renormalized candidate probabilities as its gating vector, weight calibration scenes by them, and take a query-specific weighted quantile of unpriced Global-SNSA scores. FacCP instead learns from the entire scene context and trains its embedding with the energy score so that retrieved errors match each candidate's residual law. It therefore adapts the shape and width of the tube itself, not the threshold alone.

We report whole-path coverage and footprint, defined as spatial area integrated over time with overlap between modes counted once, on test scenes (App. D.1). At the 95% target, FacCP reduces mean footprint by 13.7% relative to Global SNSA (586.4 vs. 679.6 $\mathrm{m}^2\,\mathrm{s}$) at 94.92% vs. 95.21% coverage (Table 1). With the same pricing rule, localization reduces mean footprint across different target coverages (App. D.7). MoE-CP covers 95.39% but requires 764.1 $\mathrm{m}^2\,\mathrm{s}$: reweighting the threshold cannot change the geometry, and each local threshold rests on fewer effective calibration scenes. FORCE-OPT is smallest at 197.6 $\mathrm{m}^2\,\mathrm{s}$, but one candidate tube covers the complete path in only 62.29% of scenes. Footprint is meaningful only if the target empirical coverage is reached.

Fig. 5 compares five methods that construct uncertainty sets around the same fixed MTR predictions. In the intersection scene (top row), FacCP covers the complete path with 281.0 $\mathrm{m}^2\,\mathrm{s}$, against 506.8 for Global SNSA and 522.0 for MoE-CP; in the sharp turn ($> 45°$, bottom row), it does so with 684.7 against 882.2 and 899.6. FORCE-OPT and the raw MTR head miss both paths. In both scenes the forecaster's error is largest along the direction of travel, which rotates as the path bends. Global SNSA pools errors from all reference scenes, mostly straight driving, so its ellipses keep one orientation and must widen to cover the turn. FacCP retrieves errors from scenes with similar turning candidates, so its ellipses rotate with the path and stay narrow across it.

6 Conclusion

We establish that approximating oracle residual geometries with invariant conditional mass bounds score non-pivotality, local coverage error, and coverage loss under covariate shift. We characterize uncertainty sufficiency through excess energy-score risk and use the proper energy score to learn residual-law retrieval. FacCP constructs multiple set geometries from retrieved laws without retraining, while a single pooled conformal quantile preserves finite-sample marginal validity. Experiments validate the geometry–pivotality connection and demonstrate tighter high-dimensional reachability sets and smaller driving prediction tubes, particularly under limited data and model capacity.

Limitations. Our local coverage bounds depend on the population pivotality defect, which we assess empirically but do not certify with finite-sample bounds. Our covariate shift guarantees also require an unchanged conditional response law and do not extend to concept shift. Developing finite-sample pivotality certificates and guarantees under concept shift remains open.

AI USE STATEMENT

Generative AI tools were used in the preparation of this work to assist with refining mathematical statements and annotations, providing feedback on human-designed experimental methodology and ablation, and improving the organization, clarity, and presentation of the manuscript. They were also used for language editing, restructuring sections, and formatting equations and tables.

All AI-assisted mathematical arguments were reviewed and revised by the authors, and all reported experimental results were produced and verified by the authors. Generative AI was not treated as a source of scientific evidence or authority; technical claims, derivations, and conclusions were checked against the underlying analysis and experimental results. The authors take full responsibility for the final content of the paper, including all text, claims, and artifacts produced with the assistance of generative AI.

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A PROOFS

Assumption 5 (Oracle common-coordinate representation)**.** *There exists a probability law Q_0 on a canonical residual-coordinate space $\mathcal{C}$ and measurable maps $\Phi_x : \mathcal{R} \to \mathcal{C}$, indexed by $x \in \mathcal{X}_0$, such that $(\Phi_x)_{\#} P_x^R = Q_0$ for all $x \in \mathcal{X}_0$ ($\#$ denotes pushforward).*

Fix a nested family $\{K_a\}_{a \geq 0}$ in $\mathcal{C}$ and define $\rho(u) := \inf\{a \geq 0 : u \in K_a\}$. The induced oracle geometry and score are

$$G_a^\star(x) := \{r \in \mathcal{R} : \Phi_x(r) \in K_a\}, \qquad S^\star(x, y) := \rho(\Phi_x(T(x, y))),$$

and let $F_0(a) := Q_0(K_a)$. This construction is used only for analysis; FacCP does not estimate Φ_x explicitly.

Theorem 6 (Oracle pivotality)**.** *Under Assumption 5, if $U \sim Q_0$, then $S^\star(X, Y) \mid X = x \overset{d}{=} \rho(U)$ for all $x \in \mathcal{X}_0$. Hence $S^\star$ is pivotal with respect to X.*

Proof. Conditional on $X = x$, Assumption 5 gives $\Phi_x(R) \sim Q_0$. Applying the same measurable functional ρ at every query gives $\rho(\Phi_x(R)) \overset{d}{=} \rho(U)$, whose distribution is independent of x. □

Proof of Theorem 3. By the score-set equivalence (i.e., $\widehat{S}(x, y) \leq a$ is equivalent to $T(x, y) \in \widehat{G}_a(x)$), $F_x(a) = P_x^R(\widehat{G}_a(x))$, while we also have $F_0(a) = P_x^R(G_a^\star(x))$ for all $x \in \mathcal{X}_0$ and for all $a \geq 0$. For any probability measure P and measurable sets A, B, $|P(A) - P(B)| \leq P(A \triangle B)$. Hence Assumption 2 gives $|F_x(a) - F_0(a)| \leq \varepsilon_{\text{geom}}$ uniformly in $x \in \mathcal{X}_0$ and $a \geq 0$.

Averaging $F_x(a)$ over X gives $F(a)$, so $|F(a) - F_0(a)| \leq \varepsilon_{\text{geom}}$. Therefore,

$$|F_x(a) - F(a)| \leq |F_x(a) - F_0(a)| + |F(a) - F_0(a)| \leq 2\varepsilon_{\text{geom}}, \tag{9}$$

and taking the supremum proves the result. □

Corollary 7 (Pivotality controls population local coverage)**.** *Suppose the score $\widehat{S}$ has pivotality defect $\Delta_{\text{piv}} = \sup_{x \in \mathcal{X}_0,\, a \geq 0} |F_x(a) - F(a)|$. Then, for any threshold a,*

$$\sup_{x \in \mathcal{X}_0} |F_x(a) - (1 - \alpha)| \leq \Delta_{\text{piv}} + |F(a) - (1 - \alpha)|\,. \tag{10}$$

In particular, let $q_\alpha := F^{-1}(1 - \alpha) := \inf\{a : F(a) \geq 1 - \alpha\}$, where F^{-1} denotes the generalized inverse, and suppose $F(q_\alpha) = 1 - \alpha$. Then

$$\sup_{x \in \mathcal{X}_0} \left| \mathbb{P}\Big(\widehat{S}(X, Y) \leq q_\alpha \mid X = x\Big) - (1 - \alpha) \right| \leq \Delta_{\text{piv}}. \tag{11}$$

Thus, if the exact marginal population quantile were known, the pivotality defect alone bounds the worst-case local coverage error.

Proof. For any x and threshold a, the triangle inequality gives

$$|F_x(a) - (1 - \alpha)| \leq |F_x(a) - F(a)| + |F(a) - (1 - \alpha)|\,.$$

By the definition of Δ_{piv}, $|F_x(a) - F(a)| \leq \Delta_{\text{piv}}$ uniformly over $x \in \mathcal{X}_0$ and $a \geq 0$. Taking the supremum over $x \in \mathcal{X}_0$ therefore yields

$$\sup_{x \in \mathcal{X}_0} |F_x(a) - (1 - \alpha)| \leq \Delta_{\text{piv}} + |F(a) - (1 - \alpha)|\,,$$

which proves (10).

Now set $a = q_\alpha$. Since $F(q_\alpha) = 1 - \alpha$, the second term vanishes, giving

$$\sup_{x \in \mathcal{X}_0} |F_x(q_\alpha) - (1 - \alpha)| \leq \Delta_{\text{piv}}.$$

Finally, by definition, $F_x(q_\alpha) = \mathbb{P}(\widehat{S}(X, Y) \leq q_\alpha \mid X = x)$, which gives (4). □

Theorem 8 (Local coverage under global calibration)**.** *Assume that, conditional on $\mathcal{D}_0$, the calibration observations are i.i.d. from the target distribution and that the marginal score distribution F is continuous. Let $k = \lceil (n+1)(1-\alpha) \rceil \le n$ and $\widehat{q}_\alpha = S_{(k)}$. Then, with probability at least $1-\delta$ over $\mathcal{D}_{\mathrm{cal}}$,*

$$\sup_{x \in \mathcal{X}_0} \left| \mathbb{P}\Big(Y_{\mathrm{new}} \in \widehat{C}_\alpha(x) \mid X_{\mathrm{new}} = x,\, \mathcal{D}_0,\, \mathcal{D}_{\mathrm{cal}} \Big) - (1-\alpha) \right| \le \Delta_{\mathrm{piv}} + \sqrt{\frac{\log(2/\delta)}{2n}} + \frac{2}{n}. \quad (12)$$

Proof. Let $\widehat{F}_n(a) := n^{-1} \sum_{i=1}^n \mathbf{1}\{S_i \le a\}$. By the Dvoretzky–Kiefer–Wolfowitz inequality, with probability at least $1-\delta$,

$$\sup_a \left| \widehat{F}_n(a) - F(a) \right| \le \varepsilon_n(\delta), \qquad \varepsilon_n(\delta) := \sqrt{\frac{\log(2/\delta)}{2n}}. \quad (13)$$

Continuity of F implies that the calibration scores are almost surely distinct, so $\widehat{F}_n(\widehat{q}_\alpha) = k/n$. By the definition of k, $|k/n - (1-\alpha)| \le 2/n$. Hence, on the DKW event,

$$|F(\widehat{q}_\alpha) - (1-\alpha)| \le \varepsilon_n(\delta) + \frac{2}{n}. \quad (14)$$

For every $x \in \mathcal{X}_0$, the definition of Δ_{piv} gives

$$|F_x(\widehat{q}_\alpha) - (1-\alpha)| \le \Delta_{\mathrm{piv}} + \varepsilon_n(\delta) + \frac{2}{n}. \quad (15)$$

Finally, by score-set equivalence (i.e., $\widehat{S}(x,y) \le a$ is equivalent to $T(x,y) \in \widehat{G}_a(x)$), $F_x(\widehat{q}_\alpha)$ is exactly the conditional coverage probability of $\widehat{C}_\alpha(x)$ given $\mathcal{D}_0$ and $\mathcal{D}_{\mathrm{cal}}$. □

α-local pivotality. For $\eta \ge 0$, define the near-quantile region and its pivotality defect by

$$\mathcal{T}_{\alpha,\eta} := \{a : |F(a) - (1-\alpha)| \le \eta\}, \qquad \Delta_{\alpha,\eta} := \sup_{x \in \mathcal{X}_0} \sup_{a \in \mathcal{T}_{\alpha,\eta}} |F_x(a) - F(a)|.$$

Thus, for any $a \in \mathcal{T}_{\alpha,\eta}$, Cor. 7 gives

$$\sup_{x \in \mathcal{X}_0} |F_x(a) - (1-\alpha)| \le \Delta_{\alpha,\eta} + \eta.$$

Empirically, we use

$$\widehat{\Delta}_{\alpha,\eta_n} := \max_g \max_{a:\, |\widehat{F}_{\mathrm{cal}}(a) - (1-\alpha)| \le \eta_n} |\widehat{F}_g(a) - \widehat{F}(a)|, \qquad \eta_n := \sqrt{\frac{\log(2/\delta)}{2n}} + \frac{2}{n},$$

where g ranges over reference-fit query-difficulty bins. This is a held-out finite-bin diagnostic, not a uniform conditional-coverage certificate.

Corollary 9 (Covariate-shift transfer)**.** *Let P^{s} and P^{t} satisfy $P^{\mathrm{s}}_{Y|X} = P^{\mathrm{t}}_{Y|X}$ on $\mathcal{X}_0$, containing both covariate supports. Conditional on $\mathcal{D}_0$, let $\mathcal{D}_{\mathrm{cal}}$ consist of n i.i.d. source observations, independent of the target test observation. Let $\widehat{C}_\alpha$ use the unweighted source split-conformal threshold, and let Δ_{piv} be defined under P^{s}. Then*

$$P^{\mathrm{t}}\Big(Y_{\mathrm{new}} \in \widehat{C}_\alpha(X_{\mathrm{new}}) \mid \mathcal{D}_0 \Big) \ge 1 - \alpha - \Delta_{\mathrm{piv}}. \quad (16)$$

Under Assumption 2, $\Delta_{\mathrm{piv}} \le 2\varepsilon_{\mathrm{geom}}$.

Furthermore, if the source marginal score CDF F^{s} is continuous and $k = \lceil (n+1)(1-\alpha) \rceil \le n$, then, with probability at least $1-\delta$ over $\mathcal{D}_{\mathrm{cal}}$,

$$\sup_{x \in \mathcal{X}_0} \left| P^{\mathrm{t}}\Big(Y_{\mathrm{new}} \in \widehat{C}_\alpha(x) \mid X_{\mathrm{new}} = x, \mathcal{D}_0, \mathcal{D}_{\mathrm{cal}} \Big) - (1-\alpha) \right| \le \Delta_{\mathrm{piv}} + \sqrt{\frac{\log(2/\delta)}{2n}} + \frac{2}{n}. \quad (17)$$

The same bound holds for the target marginal coverage error conditional on $\mathcal{D}_0, \mathcal{D}_{\mathrm{cal}}$.

Proof. Condition on $\mathcal{D}_0$. Let F_x^{s} and F_x^{t} denote the source and target conditional score CDFs. Since the conditional response law is preserved, $F_x^{\mathrm{t}}(a) = F_x^{\mathrm{s}}(a)$ for every $x \in \mathcal{X}_0$ and a. Consequently,

$$|F^{\mathrm{t}}(a) - F^{\mathrm{s}}(a)| = \left| \int_{\mathcal{X}_0} \big(F_x^{\mathrm{s}}(a) - F^{\mathrm{s}}(a)\big)\, dP_X^{\mathrm{t}}(x) \right| \le \Delta_{\mathrm{piv}}.$$

Conditional on $\mathcal{D}_{\mathrm{cal}}$, the threshold $\widehat{q}_\alpha$ is fixed, and target coverage has the equivalent geometric representation

$$P^{\mathrm{t}}\Big(Y_{\mathrm{new}} \in \widehat{C}_\alpha(X_{\mathrm{new}}) \mid \mathcal{D}_0, \mathcal{D}_{\mathrm{cal}}\Big) = \int_{\mathcal{X}_0} P_x^R\Big(\widehat{G}_{\widehat{q}_\alpha}(x)\Big)\, dP_X^{\mathrm{t}}(x) = F^{\mathrm{t}}(\widehat{q}_\alpha).$$

Thus

$$F^{\mathrm{t}}(\widehat{q}_\alpha) \ge F^{\mathrm{s}}(\widehat{q}_\alpha) - \Delta_{\mathrm{piv}}.$$

Taking expectations over $\mathcal{D}_{\mathrm{cal}}$ and applying the source split-conformal guarantee from Theorem 10 gives

$$P^{\mathrm{t}}\Big(Y_{\mathrm{new}} \in \widehat{C}_\alpha(X_{\mathrm{new}}) \mid \mathcal{D}_0\Big) \ge \mathbb{E}[F^{\mathrm{s}}(\widehat{q}_\alpha) \mid \mathcal{D}_0] - \Delta_{\mathrm{piv}} \ge 1 - \alpha - \Delta_{\mathrm{piv}}.$$

For the finite-calibration refinement, the DKW argument in Theorem 8 gives, with probability at least $1 - \delta$,

$$|F^{\mathrm{s}}(\widehat{q}_\alpha) - (1-\alpha)| \le \sqrt{\frac{\log(2/\delta)}{2n}} + \frac{2}{n}.$$

Since $F_x^{\mathrm{t}} = F_x^{\mathrm{s}}$,

$$\begin{aligned} |F_x^{\mathrm{t}}(\widehat{q}_\alpha) - (1-\alpha)| &\le |F_x^{\mathrm{s}}(\widehat{q}_\alpha) - F^{\mathrm{s}}(\widehat{q}_\alpha)| + |F^{\mathrm{s}}(\widehat{q}_\alpha) - (1-\alpha)| \\ &\le \Delta_{\mathrm{piv}} + \sqrt{\frac{\log(2/\delta)}{2n}} + \frac{2}{n}. \end{aligned}$$

Taking the supremum over $\mathcal{X}_0$ proves the conditional bound. Integrating over P_X^{t} gives the corresponding target marginal bound. □

This perspective is related to Gibbs et al. (2025), who formulate conditional validity as coverage over classes of covariate shifts and calibrate against specified classes of reweightings. Rather than adapting calibration to a target shift, FacCP seeks an approximately pivotal score whose conditional law is invariant in x. When this structural property holds over the target-relevant region, the same unweighted source threshold transfers without target-specific density-ratio estimation; when it fails, shift-aware calibration remains necessary (Tibshirani et al., 2019; Barber et al., 2023; Joshi et al., 2026).

Theorem 10 (Finite-sample marginal validity)**.** *Let $\mathcal{D}_0$ denote all data and randomness used to define $\widehat{S}$. Assume that, conditional on $\mathcal{D}_0$, $(X_1, Y_1), \ldots, (X_n, Y_n), (X_{\mathrm{new}}, Y_{\mathrm{new}})$ are exchangeable. Then*

$$\mathbb{P}\Big(Y_{\mathrm{new}} \in \widehat{C}_\alpha(X_{\mathrm{new}}) \mid \mathcal{D}_0\Big) \ge 1 - \alpha, \tag{18}$$

$$\mathbb{P}\Big(Y_{\mathrm{new}} \in \widehat{C}_\alpha(X_{\mathrm{new}})\Big) \ge 1 - \alpha. \tag{19}$$

Proof. Conditional on $\mathcal{D}_0$, the score function $\widehat{S} : \mathcal{X} \times \mathcal{Y} \to [0, \infty]$ is fixed. Since $(X_1, Y_1), \ldots, (X_n, Y_n), (X_{\mathrm{new}}, Y_{\mathrm{new}})$ are exchangeable conditional on $\mathcal{D}_0$, applying the same fixed function $\widehat{S}$ to each observation implies that the scores $S_1, \ldots, S_n, S_{\mathrm{new}}$, where $S_{\mathrm{new}} = \widehat{S}(X_{\mathrm{new}}, Y_{\mathrm{new}})$, are also exchangeable conditional on $\mathcal{D}_0$.

Let $k = \lceil (n+1)(1-\alpha) \rceil$. By exchangeability, the rank of S_{new} among the $n+1$ scores is uniform up to ties. Hence the split-conformal order statistic satisfies

$$\mathbb{P}\big(S_{\mathrm{new}} \le S_{(k)} \mid \mathcal{D}_0\big) \ge \frac{k}{n+1} \ge 1 - \alpha. \tag{20}$$

Here $S_{(k)}$ denotes the kth order statistic of the calibration scores, with the convention $S_{(n+1)} = \infty$.

By definition of the conformal prediction set, $\widehat{C}_\alpha(x) = \{y \in \mathcal{Y} : \widehat{S}(x,y) \le S_{(k)}\}$, so the event $S_{\text{new}} \le S_{(k)}$ is exactly $Y_{\text{new}} \in \widehat{C}_\alpha(X_{\text{new}})$. Equivalently, since $\widehat{S}(x,y) \le a$ if and only if $T(x,y) \in \widehat{G}_a(x)$, this event is the same as $T(X_{\text{new}}, Y_{\text{new}}) \in \widehat{G}_{S_{(k)}}(X_{\text{new}})$. Therefore,

$$\mathbb{P}\Big(Y_{\text{new}} \in \widehat{C}_\alpha(X_{\text{new}}) \mid \mathcal{D}_0\Big) \ge 1 - \alpha, \tag{21}$$

which proves (18).

Finally, by the law of total probability, $\mathbb{P}(A) = \mathbb{E}[\mathbb{P}(A \mid \mathcal{D}_0)]$. Applying this to the event $A = \{Y_{\text{new}} \in \widehat{C}_\alpha(X_{\text{new}})\}$ and using the conditional bound above gives

$$\mathbb{P}\Big(Y_{\text{new}} \in \widehat{C}_\alpha(X_{\text{new}})\Big) = \mathbb{E}\Big[\mathbb{P}\Big(Y_{\text{new}} \in \widehat{C}_\alpha(X_{\text{new}}) \mid \mathcal{D}_0\Big)\Big] \ge 1 - \alpha, \tag{22}$$

proving (19). □

Proof of Proposition 4. Conditional on X, $R \sim P_X^R$, while P_Z^R is fixed because $Z = h_\theta(X)$. Applying the energy-score identity conditionally on X gives

$$\mathbb{E}\big[\mathrm{ES}(P_Z^R, R) - \mathrm{ES}(P_X^R, R) \mid X\big] = \frac{1}{2}\mathcal{E}(P_X^R, P_Z^R).$$

Taking expectations over X proves the stated identity. Since energy distance is nonnegative and vanishes exactly when its arguments coincide, equality of the risks is equivalent to $P_X^R = P_Z^R$ almost surely, which is precisely uncertainty sufficiency. □

Proposition 11 (Retrieval loss is the energy score)**.** *For the retrieved residual law* $\widehat{P}^R_{x_i,\theta} = \sum_{j\in J_i} w_{ij}^{(\theta)} \delta_{r_j}$*, the training loss in* (8) *satisfies* $\ell_i(\theta) = \mathrm{ES}(\widehat{P}^R_{x_i,\theta}, r_i)$.

Proof. If $V \sim \widehat{P}^R_{x_i,\theta}$, then $V = r_j$ with probability $w_{ij}^{(\theta)}$. Hence

$$\mathbb{E}\|V - r_i\|_2 = \sum_{j\in J_i} w_{ij}^{(\theta)} \|r_j - r_i\|_2.$$

Likewise, for independent $V, V' \sim \widehat{P}^R_{x_i,\theta}$,

$$\mathbb{E}\|V - V'\|_2 = \sum_{j,k\in J_i} w_{ij}^{(\theta)} w_{ik}^{(\theta)} \|r_j - r_k\|_2.$$

Substituting these expressions into the definition of the energy score yields

$$\mathrm{ES}\Big(\widehat{P}^R_{x_i,\theta}, r_i\Big) = \sum_{j\in J_i} w_{ij}^{(\theta)} \|r_j - r_i\|_2 - \frac{1}{2} \sum_{j,k\in J_i} w_{ij}^{(\theta)} w_{ik}^{(\theta)} \|r_j - r_k\|_2,$$

which is exactly $\ell_i(\theta)$. □

For the following result, let U denote the variable used for localization and use the kernel-weighted estimator in (6), with weights proportional to $\exp(-d_U(U_j, u)/\tau)$. Define the energy metric

$$d_{\mathrm{E}}(P, Q) := \left(\frac{\mathcal{E}(P, Q)}{2}\right)^{1/2}, \tag{23}$$

so that $d_{\mathrm{E}}^2(P, Q)$ equals the excess expected energy-score risk incurred by using Q when the true law is P.

Proposition 12 (Penalty from uncertainty-irrelevant dimensions)**.** *Let* $S = g(X) \in \mathbb{R}^d$ *be uncertainty-sufficient, and let* $U = (S, N) \in \mathbb{R}^p$*, where* $N = n(X) \in \mathbb{R}^{p-d}$*,* $p > d$*, and* $R \perp N \mid S$*. Fix a query with coordinates* $s_0 \in \mathrm{supp}(P_S)$ *and* $u_0 = (s_0, n_0) \in \mathrm{supp}(P_U)$*. Assume that, in neighborhoods of these points, the retrieval geometries are regular and equivalent to Euclidean geometry, and that* S *and* U *have continuous densities bounded above and below by positive constants. Assume also that* $P_s^R := \mathcal{L}(R \mid S = s)$ *is locally* β*-Hölder in energy metric: for*

some $r > 0$, $d_{\mathrm{E}}(P_s^R, P_{s_0}^R) \le L\|s - s_0\|^\beta$, for all $s \in \mathrm{supp}(P_S) \cap B(s_0, r)$, with $0 < \beta \le 1$, and that $\mathcal{R}$ is bounded.

For the kernel estimator in (6)*, assume the isotropic kernel is bounded, nonnegative, nonzero near the origin, and has finite moments sufficient for the expressions below. Let $\mathfrak{P}_{\beta,L}$ denote the resulting class of distributions, and define $\mathfrak{R}_d(\tau) := \sup_{P\in\mathfrak{P}_{\beta,L}} \mathbb{E}_P[\mathcal{E}(\widehat{P}_{S,\tau}^R, P_{s_0}^R)]$ and $\mathfrak{R}_p(\tau) := \sup_{P\in\mathfrak{P}_{\beta,L}} \mathbb{E}_P[\mathcal{E}(\widehat{P}_{U,\tau}^R, P_{s_0}^R)]$. Then, as $\tau \to 0$ and $m\tau^p \to \infty$, where $f \asymp g$ means $cg \le f \le Cg$ for some constants $0 < c \le C < \infty$,*

$$\mathfrak{R}_d(\tau) \asymp \tau^{2\beta} + \frac{1}{m\tau^d}, \qquad \mathfrak{R}_p(\tau) \asymp \tau^{2\beta} + \frac{1}{m\tau^p}. \tag{24}$$

If $\tau_d^\star \in \arg\min_\tau \mathfrak{R}_d(\tau)$ and $\tau_p^\star \in \arg\min_\tau \mathfrak{R}_p(\tau)$, then $\tau_d^\star \asymp m^{-1/(2\beta+d)}$ and $\tau_p^\star \asymp m^{-1/(2\beta+p)}$, and

$$\mathfrak{R}_d(\tau_d^\star) \asymp m^{-2\beta/(2\beta+d)}, \qquad \mathfrak{R}_p(\tau_p^\star) \asymp m^{-2\beta/(2\beta+p)}. \tag{25}$$

Thus, although N contains no additional information about the residual law, allowing it to vary under the retrieval metric slows the worst-case localization rate whenever $p > d$.

Proof. We prove the result for a generic localization variable V of local dimension $s \in \{d, p\}$ and then substitute $V = S$ and $V = U$.

Let $\mu(s) := \mathbb{E}[\phi(R) \mid S = s]$ denote the conditional mean embedding associated with energy distance, so that $d_{\mathrm{E}}(P, Q) = \|\mu(P) - \mu(Q)\|_{\mathcal{H}}$ and $\mathcal{E}(P, Q) = 2\|\mu(P) - \mu(Q)\|_{\mathcal{H}}^2$. Writing the weighted empirical embedding as $\widehat{\mu}_{V,\tau} := \sum_i w_i^{V,\tau}\phi(R_i)$, we have

$$\widehat{\mu}_{V,\tau} - \mu(s_0) = \sum_i w_i^{V,\tau}\big(\phi(R_i) - \mu(S_i)\big) + \sum_i w_i^{V,\tau}\big(\mu(S_i) - \mu(s_0)\big).$$

Conditional on the reference locations, the first term has mean zero, so the cross term vanishes after expectation. Hence the expected squared embedding error separates into a finite-reference variance term and a squared localization-bias term.

For the bias, kernel localization at scale τ assigns asymptotically all effective weight to points whose retrieval distance from the query is $O(\tau)$. Local regularity of the retrieval geometry therefore implies $\|S_i - s_0\| = O(\tau)$ for effectively weighted samples. By the Hölder assumption, $\|\mu(S_i) - \mu(s_0)\|_{\mathcal{H}} = O(\tau^\beta)$, and thus the squared bias is $O(\tau^{2\beta})$.

For the variance, boundedness of $\mathcal{R}$ implies $\mathbb{E}[\|\phi(R) - \mu(S)\|_{\mathcal{H}}^2 \mid S] \le C$ for some finite C. Conditional independence therefore gives

$$\mathbb{E}\left[\left\|\sum_i w_i^{V,\tau}\big(\phi(R_i) - \mu(S_i)\big)\right\|_{\mathcal{H}}^2 \,\middle|\, V_1, \ldots, V_m\right] \le C\sum_i (w_i^{V,\tau})^2.$$

Because the local density of V is bounded above and below, standard kernel-sum concentration gives $\sum_i (w_i^{V,\tau})^2 = \Theta_p((m\tau^s)^{-1})$ and therefore an expected variance of order $(m\tau^s)^{-1}$. This proves the upper bound

$$\sup_{P\in\mathfrak{P}_{\beta,L}} \mathbb{E}_P[\mathcal{E}(\widehat{P}_{V,\tau}^R, P_{s_0}^R)] = O\left(\tau^{2\beta} + \frac{1}{m\tau^s}\right).$$

It remains to establish matching lower orders. For the variance term, restrict $\mathfrak{P}_{\beta,L}$ to a locally constant, nondegenerate two-point residual distribution. The localization bias then vanishes, while the conditional embedding variance is bounded below by a positive constant. The same kernel-weight calculation gives

$$\sup_{P\in\mathfrak{P}_{\beta,L}} \mathbb{E}_P[\mathcal{E}(\widehat{P}_{V,\tau}^R, P_{s_0}^R)] = \Omega\left(\frac{1}{m\tau^s}\right).$$

For the bias term, again restrict to residuals supported on two distinct points, but let the probability of one point vary locally as $\pi(s) = \pi_0 + c\|s - s_0\|^\beta$, with $c > 0$ chosen small enough that $\pi(s) \in (0, 1)$. This family lies in the assumed Hölder class. For two-point laws, energy distance is a positive constant times the squared difference in their probabilities. Local kernel averaging at scale τ changes

$\pi(s_0)$ by order τ^β, because the local density is positive and the kernel assigns nonvanishing mass to points at distance proportional to τ. Hence

$$\sup_{P\in\mathfrak{P}_{\beta,L}} \mathbb{E}_P[\mathcal{E}(\widehat{P}^R_{V,\tau}, P^R_{s_0})] = \Omega(\tau^{2\beta}).$$

Combining the two lower bounds and using $\max\{a,b\} \geq (a+b)/2$ yields

$$\sup_{P\in\mathfrak{P}_{\beta,L}} \mathbb{E}_P[\mathcal{E}(\widehat{P}^R_{V,\tau}, P^R_{s_0})] \asymp \tau^{2\beta} + \frac{1}{m\tau^s}.$$

Taking $s = d$ and $s = p$ proves (24). Finally, minimizing $\tau^{2\beta} + (m\tau^s)^{-1}$ gives $\tau^\star_s \asymp m^{-1/(2\beta+s)}$ and optimized risk $m^{-2\beta/(2\beta+s)}$, yielding (25). □

Interpretation. The p-dimensional penalty applies only when the additional coordinates introduce genuine local variation under the retrieval metric. If a p-dimensional representation is merely a redundant embedding of a d-dimensional manifold, its local probability mass still scales according to dimension d, and the p-dimensional penalty does not apply. Thus, the relevant quantity is the intrinsic localization dimension rather than the number of stored coordinates.

To see the statistical mechanism, define the effective local support

$$N_{\text{eff}}(u,\tau) := \frac{1}{\sum_j w_j(u)^2}. \tag{26}$$

At a fixed localization scale, uncertainty-irrelevant directions can separate reference samples under the retrieval metric without improving approximation of the conditional residual law, thereby reducing N_{eff}. Maintaining the same effective support then requires broader localization, which increases bias. An ideal uncertainty-relevant representation collapses such nuisance directions, whereas representation-learning error may leave variation along them and increase the empirical localization dimension.

This benefit must be balanced against uncertainty sufficiency. Excessive compression can discard information that genuinely changes P^R_x, incurring the representation error characterized in Prop. 4. The desired representation therefore preserves uncertainty-relevant variation while suppressing nuisance variation that makes residual-law retrieval statistically harder.

B Additional Related Work

Adaptive scores, pivotality, and conditional coverage. Standard CP provides finite-sample marginal coverage under exchangeability (Vovk et al., 2005; Shafer & Vovk, 2008; Lei et al., 2018). Exact distribution-free pointwise conditional coverage is generally impossible without additional assumptions (Lei & Wasserman, 2014; Barber et al., 2021); Vovk (2012) distinguishes conditional validity criteria, while Cauchois et al. (2021) studies adaptive and validated sets for multiclass and multilabel outcomes. Our theoretical contribution is a geometry-specific sufficient condition for approximate pivotality. If the learned sublevel sets uniformly disagree with oracle sets of query-invariant conditional mass by at most $\varepsilon_{\text{geom}}$ in conditional probability, the resulting score's pivotality defect is at most $2\varepsilon_{\text{geom}}$. This statement applies to any suitable learned geometry, not only to FacCP, and separates geometric approximation error from finite calibration error.

Early normalized nonconformity methods adapt regression intervals using estimates of local difficulty (Papadopoulos et al., 2008; Papadopoulos & Haralambous, 2011). CQR learns response quantiles and conformalizes their asymmetric intervals (Romano et al., 2019); self-supervised adaptation (Seedat et al., 2023), quantile-forest-based score reweighting (Amoukou & Brunel, 2023), and adaptive Split-Jackknife+ scores (Deutschmann et al., 2024) likewise modify the way prediction errors are scored or calibrated. These approaches express their adaptation through a selected scalar score or interval family. They do not, by score adjustment alone, recover multivariate orientation.

Two recent approaches address score pivotality through post-hoc transformations rather than through the construction of local geometry. Laplante (2026) propose PIT-CP, which estimates the conditional CDF of a given score and transforms it toward a query-invariant distribution. While this can achieve

approximate pivotality, the transformation only reindexes the base score's nested sets and cannot correct misspecified geometry. Their symmetric-difference guarantees compare corrected sets to oracle conditional sets within that same fixed geometric family, rather than relating geometric approximation to score pivotality. Similarly, Plassier et al. (2025a) rectify an existing score using estimated conditional quantiles, targeting conditional coverage at a specified level without establishing uniform score pivotality. In contrast, our result provides a direct sufficient condition for pivotality from geometry. Uniformly approximating oracle geometric sublevel sets with query-invariant conditional probability mass bounds the pivotality defect of the induced score simultaneously across all inflation levels. Thus, rather than estimating conditional score distributions or quantiles to correct non-pivotality after scoring, our approach constructs approximately pivotal scores through local geometry estimation, followed by a single globally pooled conformal calibration. Crucially, score rectification can improve conditional coverage even with misspecified geometry, but cannot recover geometric structure absent from the base score.

Other approaches target score homogeneity at a different stage. Colombo (2024) learn input-dependent monotone transformations of scalar conformity scores through normalizing flows. Gao et al. (2024) adjust the regression predictor itself using discrepancies between conditional and marginal score distributions. The latter objective closely resembles the pivotality defect, but it changes the predictor to improve a chosen score. In contrast, FacCP constructs query-dependent geometric sublevel sets directly, and establishes that approximating oracle geometries with invariant conditional probability mass yields approximately pivotal scores. The distinction is between transforming or regularizing scores after choosing their underlying geometry and using learned geometric sublevel sets as a sufficient route to score comparability. Hanselle et al. (2025) additionally show that ordinary conformal inference can be expressed through score orderings without explicit real-valued scores; approximate pivotality asks for the stronger property that common score thresholds also have comparable conditional mass.

Localized and group-conditional conformal prediction. Localized CP adapts the calibration rule to the query. Guan (2023) use similarity-dependent calibration, and randomized localized CP (Hore & Barber, 2025) supplies local weighting with finite-sample guarantees for relaxed notions of conditional coverage. Such methods can correct local undercoverage for a given score, but changing its calibration threshold alone does not introduce geometric distinctions absent from its sublevel sets. Their principal statistical tradeoff is that a query-dependent tail threshold may be informed by fewer effective calibration observations than a pooled threshold. FacCP instead pools calibration scores after incorporating query dependence into the score geometry. Global calibration remains marginally valid regardless of retrieval accuracy, while its local reliability depends on approximate score pivotality.

Some localized approaches already modify scores upstream of final calibration. Han et al. (2022) construct a localized score using kernel estimates of conditional information and retain a split-conformal procedure. FacCP differs in learning the localization metric by evaluating the entire weighted residual distribution that the metric retrieves. Kiyani et al. (2024) learn uncertainty-guided features and partitions for conditional calibration in PLCP, whereas FacCP's representation is trained to preserve information about the conditional residual law rather than to define calibration groups. These objectives produce different intermediate quantities. A partition or locally weighted score need not estimate a multivariate residual law that can be reused by several geometry constructors. Related approaches based on mixture-of-experts gating similarities (Kong et al., 2026) also learn how to weight calibration examples.

Pooling observations with similar uncertainty has precedents outside query-based regression. Ding et al. (2023) group classes with related score distributions to increase support for class-conditional calibration. FacCP similarly exploits common uncertainty structure, but pools residual observations for pre-calibration law estimation rather than pooling calibration observations across output classes. More generally, Gibbs et al. (2025) formulate conditional guarantees over specified classes of covariate reweightings. FacCP does not directly optimize coverage against such a class. It aims to construct a score whose conditional law is stable over a specified covariate region, so that one unweighted threshold has approximately common conditional meaning there. These approaches address different assumptions and coverage objectives.

Conditional-distribution and residual-law modeling. Conditional distribution estimates can be used to define adaptive conformal scores and regions. COPS (Lei & Wasserman, 2014), density-based split methods (Izbicki et al., 2020; 2022), and distributional CP (Chernozhukov et al., 2021) use estimated conditional densities, density levels, or distribution functions. SPICE learns conditional spline densities (Diamant et al., 2024). Probabilistic conformal prediction constructs regions from conditional generative samples and can represent disconnected output sets (Wang et al., 2023). These methods need not restrict uncertainty to a scalar scale or a single ellipsoid. Their difference from FacCP is the conditional estimation mechanism. FacCP learns query-space coordinates for retrieving observed residuals, yielding a weighted empirical distribution in the original residual space without fitting a conditional density or generator.

CP2 (Plassier et al., 2025b) is the closest distributional comparison and already supports query-adaptive nested regions with globally pooled calibration. It also derives nonasymptotic conditional-coverage bounds using the total-variation discrepancy between a fitted conditional response distribution and the true one. Our analysis controls a different object. For oracle geometries $G_a^\star(x)$ with conditional mass independent of x, it bounds the discrepancy of the induced score laws through $P_x^R(\widehat{G}_a(x) \triangle G_a^\star(x))$, uniformly over queries and inflation levels. Total variation bounds errors on arbitrary measurable events, whereas the geometry condition evaluates disagreement on the particular sublevel sets defining the score. The assumptions are distinct sufficient conditions, not a general ordering of weaker and stronger requirements. In particular, neither the strict propriety of the retrieval objective nor consistency of a conditional-law estimator alone guarantees the requisite set approximation without stability of the geometry constructor. The geometry-to-pivotality statement can also apply to covariance, template, and transformation models that do not fit a normalized conditional law. FacCP's implementation is distinguished further by its fixed base predictor and learned residual-law retrieval mechanism.

Other full-law constructions differ in what they retain and how they form sets. Matabuena et al. (2024) estimate conditional kernel-depth structure for prediction regions in separable Hilbert spaces. Characteristic kernel embeddings can retain a complete probability law. They construct depth-based regions from conditional embeddings, whereas FacCP learns a query-space retrieval representation independent of the downstream geometry. Dheur et al. (2024) use predictive joint densities to construct conformal regions for marked temporal point processes, exploiting dependence between event times and marks. FacCP instead needs observed residuals from a fixed predictor and does not require that predictor to provide a joint density.

The energy score provides a specific reason to learn the retrieval distribution rather than a scale, covariance, or set boundary. It is a strictly proper scoring rule for distributions with finite first moments (Gneiting & Raftery, 2007). Its population risk is minimized by the true conditional residual law, and restricting the prediction to $Z = h_\theta(X)$ incurs an excess optimal risk equal to one half of the expected energy distance between the laws conditioned on X and on Z. The excess vanishes exactly when $R \perp X \mid Z$. Our contribution is to use strict propriety to characterize uncertainty sufficiency in the representation and to learn a query-space retrieval metric through the energy score of the weighted empirical residual law. Under the additional localization assumptions analyzed in the paper, removing residual-irrelevant coordinates can improve estimation rates.

Learned multivariate geometries and transformations. Adaptive geometric CP methods directly model properties of multivariate prediction regions. Messoudi et al. (2022) use query-dependent ellipsoids, and Xu et al. (2024) construct ellipsoidal regions for multidimensional sequential data. Both exploit covariance structure, but their geometric representation remains ellipsoidal; the latter also addresses nonexchangeable time-series calibration rather than the split-conformal residual-retrieval problem studied here. Feldman et al. (2023) learn output representations to facilitate multi-output quantile regression, whereas FacCP learns a representation of the query used to select reference residuals. Tumu et al. (2024) optimize convex shape templates for multimodal regions, and Braun et al. (2026) optimize multivariate covering sets for volume through norm-based constructions. These methods explicitly optimize geometric objects or their parameters. FacCP instead learns the residual-law retrieval mechanism and then chooses a geometry constructor, so ellipsoids, hyperboxes, zonotopes, and density-based sets can share a trained retrieval representation.

Flow and transport methods provide particularly important alternatives to simple covariance models. CONTRA learns a normalizing-flow transformation whose latent scores define conformal regions;

resCONTRA applies the construction to residuals of a predictive model (Fang et al., 2025). JAPAN uses flow-based density estimates with several conformity scores to form adaptive, potentially disconnected regions (English & Lippert, 2026). Both can represent geometry beyond ellipsoids, and JAPAN is not restricted to a single density score. Their networks learn output-space transformations or densities, whereas FacCP trains a query-space retrieval metric and derives the geometry from weighted observed residuals. This different allocation of model capacity motivates the data and parameter efficiency comparison, without asserting that retrieval must outperform expressive flows as data and capacity increase. Thurin et al. (2025) develop multivariate transport ranks and flexible regions, including adaptations with asymptotic conditional-coverage properties. The transport-based scoring mechanism is distinct from FacCP's residual-law retrieval, but the general geometry-to-pivotality condition could also be applied to a suitable learned transport geometry. Comparisons of multivariate conformity scores (Dheur et al., 2025a) further support treating the choice of geometry as consequential rather than interchangeable. Flow-based conformal predictive distributions (Harris, 2026) address sampling and representing sets induced by scores, a complementary problem to learning the local residual law used to define those scores.

Finally, weighted CP handles covariate shift by changing calibration weights (Tibshirani et al., 2019; Barber et al., 2023; Joshi et al., 2026). If the conditional response law remains fixed and the score is approximately pivotal over both source- and target-relevant covariates, FacCP's unweighted threshold can transfer with a local error governed by the pivotality defect. Diagnostics for conditional coverage (Braun et al., 2025; Cauchois et al., 2021) are complementary because marginal coverage alone cannot establish score comparability or local reliability.

C Additional Experimental Details

C.1 Controlled Residual-Geometry Experiment

Data-generating process. Let $X = (X_1, X_2, X_3, N)$, where $X_1, X_2, X_3 \overset{\text{i.i.d.}}{\sim} \mathrm{Unif}[-1, 1]$ and $N \sim \mathcal{N}(0, I_{d_{\text{nuis}}})$ is independent. Residuals follow

$$R \mid X = x \sim \mathcal{N}(0, \Sigma(x)), \qquad x = (x_1, x_2, x_3, n),$$

with

$$s(x) = \exp(0.65 x_1), \qquad a(x) = \exp\left\{ \frac{\log 8}{2}(x_2 + 1) \right\}, \qquad \theta(x) = \frac{\pi}{2} x_3,$$

and

$$\Sigma(x) = Q_{\theta(x)} \begin{bmatrix} s(x)^2 a(x) & 0 \\ 0 & s(x)^2 / a(x) \end{bmatrix} Q_{\theta(x)}^\top.$$

Thus the first three coordinates determine the scale, aspect ratio, and orientation of the conditional residual distribution, while the nuisance coordinates change X without changing P_x.

We use independent samples of size $50{,}000$ for representation training, $30{,}000$ for residual retrieval, $8{,}000$ for hyperparameter tuning, $10{,}000$ for conformal calibration, and $10{,}000$ for evaluation. The target miscoverage is 5%. All learned components and retrieval hyperparameters are fixed before conformal calibration.

Geometry constructions. Oracle geometry uses the exact conditional covariance $\Sigma(x)$ in the Mahalanobis score. Its population score distribution is invariant in x. Small nonzero empirical pivotality and coverage errors therefore arise from finite calibration and Monte Carlo evaluation.

Global geometry estimates one residual covariance from the full reference sample and uses it at every query. Only its final conformal radius changes, so it cannot represent query-dependent aspect ratio or orientation.

Privileged retrieval uses

$$\psi(x) = \left(\log s(x), \log a(x) \cos 2\theta(x), \log a(x) \sin 2\theta(x)\right).$$

The doubled angle encodes the π-periodicity of ellipse orientation. These coordinates are computed from the known data-generating factors, but they do not provide $\Sigma(x)$. The method uses the same finite reference sample, kernel retrieval, and weighted covariance construction as the other retrieval

methods. It therefore isolates the effect of localizing in coordinates that contain the relevant uncertainty factors without nuisance variation.

Raw-X retrieval uses the same kernel retrieval and covariance construction directly in the observed input space. When $\text{nuisance dimensions} = 0$, X is low-dimensional and its three coordinates determine P_x. Increasing $\text{nuisance dimensions}$ adds coordinates that affect retrieval distances without changing the conditional residual distribution.

FacCP replaces the prescribed localization coordinates with a learned energy-score representation. The representation has dimension 12 and hidden width 64 and is trained for 150 epochs. The retrieved residuals define a weighted second moment, which is used as the local ellipsoidal geometry.

Covariance prediction uses a width-64 neural network to predict a positive-definite 2×2 covariance directly from X and is trained using Gaussian negative log likelihood. Since the conditional residual distribution in this experiment is Gaussian and the target sets are ellipsoidal, this baseline is well specified.

Retrieval and calibration. For each retrieval representation $z(x)$, reference coordinates are standardized using the reference sample, except for the learned representation whose metric is fixed by training. Retrieval weights are proportional to

$$w_j(x) \propto \exp\left(-\frac{\|\widetilde{z}(x) - \widetilde{z}(X_j)\|_2^2}{\tau}\right).$$

The resulting covariance is the weighted residual second moment with relative ridge 10^{-4}. The temperature τ is selected on the independent tuning split by Gaussian negative log likelihood among candidates with mean effective sample size at least 100. A single split-conformal radius is then estimated using all $10{,}000$ calibration observations.

α-local pivotality diagnostic. We use an empirical diagnostic from Cor. 7. For each of scale, aspect ratio, and orientation, we form 10 equal-frequency bins using boundaries estimated from the reference sample. Within each held-out evaluation bin, we compare the conditional empirical score CDF with the pooled evaluation CDF over calibration thresholds satisfying

$$\left|\widehat{F}_{\text{cal}}(t) - 0.95\right| \le \eta_n, \qquad \eta_n = \sqrt{\frac{\log(2/0.05)}{2n_{\text{cal}}}} + \frac{2}{n_{\text{cal}}}.$$

We report the largest discrepancy across bins and eligible thresholds. This is a finite-bin empirical approximation to α-local pivotality and is not an estimate of the theoretical $\sup_x$ quantity.

For this controlled experiment we also report a stronger full-CDF pivotality diagnostic. We evaluate a grid of queries varying scale, aspect ratio, and orientation separately, draw fresh residuals from the known conditional distribution at every grid point, and compute the largest KS distance between each conditional score distribution and the pooled score distribution. This diagnostic evaluates the whole score CDF and is therefore stronger than the fixed-miscoverage quantity used in the main experiments.

Table 2: Controlled experiment without nuisance coordinates. Area is normalized by the oracle geometry.

Method	α-local defect	Full-CDF defect	Mean area / oracle
Oracle geometry	0.020	0.017	1.02
Global geometry	0.139	0.361	3.95
Privileged retrieval	0.020	0.074	1.06
Raw-X retrieval	0.031	0.123	1.05
FacCP	0.029	0.117	1.07
Covariance prediction	0.022	0.044	1.04

Nuisance-coordinate stress test. We repeat the experiment with $\text{nuisance dimensions} \in \{0, 4, 8, 16, 32, 64\}$ while keeping the signal coordinates and residual draws paired across nuisance

dimensions. The added coordinates never enter $\Sigma(x)$ and therefore contain no information about P_x.

Raw-X retrieval increases from $1.09\times$ at zero nuisance dimensions to $1.52\times$, $1.72\times$, $2.04\times$, $2.52\times$, and $3.20\times$ as the nuisance dimension increases. FacCP obtains $1.07\times$, $1.08\times$, $1.11\times$, $1.14\times$, $1.15\times$, and $1.19\times$, respectively. At 16 nuisance dimensions, the worst conditional-coverage deviation is 0.301 for raw-X retrieval, 0.066 for FacCP, and 0.035 for Privileged retrieval. Panels E–H of Fig. 2 show the corresponding area growth and representative prediction sets.

Localized-quantile comparison. We separately fix the globally calibrated covariance and allow the calibration threshold to depend on the query using BaseLCP, calibrated LCP, and randomized LCP. Each method is evaluated using Privileged retrieval coordinates. This comparison uses a common subset of $2{,}000$ calibration and $2{,}000$ evaluation observations.

The unlocalized fixed score has full-CDF pivotality defect 0.361. The localized procedures reduce the post-localization defect. They nevertheless retain the same underlying family of globally oriented ellipses. To remove local-quantile estimation error from this comparison, we also compute the pointwise conditional 95% quantile of the fixed global score using fresh conditional draws. Its mean matched-coverage area remains $1.99\times$ oracle. Panels A–B of Fig. 2 show why. Local calibration can resize the fixed geometry and is effective for a large near-circular query, but cannot align the set with a tilted anisotropic residual distribution. Panels C–D show that methods which adapt the residual geometry can instead change its orientation and aspect ratio before the final conformal inflation.

C.2 Covariate-Shift Sweep

Table 3: Target coverage (%) under a shift toward larger residual scale. No target-domain recalibration is performed.

β	Global	FacCP	Raw-X	Privileged	Cov. Pred.
0.00	94.30	94.95	94.75	94.83	94.73
1.00	91.60	94.57	94.60	94.47	94.43
1.22	90.68	94.87	94.87	94.43	94.80
1.44	90.73	94.73	94.87	94.28	94.88
1.67	89.53	94.70	94.53	94.52	94.87
1.89	89.20	94.97	95.18	94.55	94.95
2.11	88.27	94.22	94.35	93.87	94.17
2.33	88.88	95.08	94.98	94.92	95.12
2.56	88.02	95.18	94.87	94.40	94.97
2.78	87.92	94.53	94.60	94.10	94.67
3.00	87.25	94.90	94.92	94.35	94.70

Table 4: Target coverage (%) under a shift toward larger residual aspect ratio. The conformal threshold remains fixed at its source-calibrated value.

β	Global	FacCP	Raw-X	Privileged	Cov. Pred.
0.00	95.12	94.78	94.77	95.02	94.87
1.00	92.12	95.35	95.58	94.73	94.48
1.22	92.33	95.55	95.77	95.30	94.93
1.44	91.17	95.17	95.42	94.55	94.33
1.67	91.12	95.50	95.85	95.03	94.42
1.89	91.27	95.75	96.02	95.27	94.82
2.11	90.35	95.65	95.80	94.58	93.90
2.33	90.42	95.97	96.15	95.05	94.33
2.56	89.97	96.10	96.22	94.98	94.67
2.78	89.27	96.08	96.30	95.12	94.77
3.00	89.22	96.38	96.33	95.27	94.60

We report the full covariate-shift sweep from Section 5.1. Calibration is always performed on the source distribution $U \sim \mathrm{Unif}([-1,1]^3)$. At test time, one uncertainty coordinate U_j is replaced by

the exponentially tilted distribution

$$p_\beta(u_j) = \frac{\exp(\beta u_j)}{\int_{-1}^{1} \exp(\beta v)\, dv}, \qquad u_j \in [-1, 1],$$

with all remaining coordinates unchanged. Thus the experiment changes P_X while preserving $P_{R|X}$. The same source-calibrated conformal threshold is used for every target shift. Tables 3–5 report target coverage, averaged over three repetitions; each repetition contains $2{,}000$ target samples. Privileged retrieval localizes using the known uncertainty-generating coordinates, while Raw-X retrieval and covariance prediction use the same baselines as in Section 5.1.

Table 5: Target coverage (%) under an orientation shift. The Global score is comparatively insensitive to this shift because its pooled geometry is approximately rotation-symmetric.

β	Global	FacCP	Raw-X	Privileged	Cov. Pred.
0.00	95.38	95.15	95.20	95.15	95.23
1.00	95.30	94.68	94.80	94.95	94.62
1.22	94.97	94.18	94.60	94.43	94.27
1.44	94.57	94.33	94.40	94.40	94.33
1.67	95.03	94.67	94.27	94.73	94.48
1.89	95.17	94.78	95.03	95.10	94.72
2.11	94.87	94.37	94.13	94.47	94.07
2.33	94.87	94.87	94.87	95.02	94.63
2.56	95.12	94.52	94.77	94.88	94.42
2.78	94.87	94.80	94.70	95.05	95.12
3.00	94.65	94.53	94.50	94.53	94.00

C.3 Planar Quadcopter experiment details

Problem and retrieval. The planar quadcopter has a 6-dimensional state and 2-dimensional control. We use $H = 100$. The retrieval query consists of the nominal state rollout and control sequence, with

$$101 \cdot 6 + 100 \cdot 2 = 806$$

input coordinates. FacCP maps this query to an 8-dimensional retrieval representation. The residual contains the 6-dimensional state error over the 100 predicted steps, giving 600 residual coordinates. For every query, the same retrieval weights in Eq. (5) define a conditional empirical residual law. The geometry constructor is then changed without retraining the representation.

Trajectory-level conformal calibration. For constructors producing a time-indexed raw score u_t, we use the single nonconformity score approach (Lindemann et al., 2025). A held-out portion of the calibration set determines positive time weights λ_t, after which each calibration trajectory has scalar score

$$S_i = \max_{1 \le t \le H} \lambda_t u_{it}.$$

The remaining calibration trajectories determine the split-conformal quantile C. The resulting trajectory is covered when

$$u_t \le \frac{C}{\lambda_t}, \qquad t = 1, \dots, H.$$

Thus one conformal event controls the complete trajectory and no union bound over time is used. For the geometry experiments we use 15% of calibration trajectories for the time normalization and the remaining 85% for the conformal quantile. Local and global methods use the same calibration procedure and differ only in whether the underlying residual geometry depends on the query.

Ellipsoids. At time t, retrieved residuals define a weighted center $c_t(x)$ and covariance

$$\Sigma_t(x) = \sum_j w_j(x) \big(r_{jt} - c_t(x)\big) \big(r_{jt} - c_t(x)\big)^\top + \epsilon I.$$

The raw score is the Mahalanobis radius

$$u_t(x, r_t) = \sqrt{\big(r_t - c_t(x)\big)^\top \Sigma_t(x)^{-1} \big(r_t - c_t(x)\big)}.$$

After trajectory-level calibration, the residual set is

$$G_t(x) = \left\{ r : \big(r - c_t(x)\big)^\top \Sigma_t(x)^{-1} \big(r - c_t(x)\big) \le \left(\frac{C}{\lambda_t}\right)^2 \right\}.$$

We use covariance ridge 10^{-3}. The global baseline applies the identical construction after pooling the geometry-bank residuals instead of retrieving them conditionally.

Hyperboxes. Retrieved residuals define a weighted center $c_t(x)$ and coordinatewise base half-widths $b_{tk}(x)$ from the 0.95 empirical quantile of the absolute residual deviations. The trajectory score is

$$S(x, r_{1:H}) = \max_{t,k} \frac{|r_{tk} - c_{tk}(x)|}{b_{tk}(x) + \epsilon}.$$

A single conformal quantile C therefore gives

$$G_t(x) = \big\{ r : |r_k - c_{tk}(x)| \le C\big(b_{tk}(x) + \epsilon\big),\ \forall k \big\}.$$

This score is already trajectory-level, so no additional time normalization is required.

KDE sets. At each time, the retrieved residuals form a weighted diagonal Gaussian KDE. We use the equivalent relative negative log-density score

$$u_t(x, r_t) = -\log \sum_j w_j(x) \exp\left(-\frac{1}{2} \left\| H_t^{-1}(r_t - r_{jt}) \right\|_2^2\right),$$

where H_t is a diagonal Silverman bandwidth with a coordinatewise floor. Since the omitted Gaussian normalizer depends only on time, this score has the same timewise density sublevel sets as the KDE. SNSA calibration gives the nonconvex set

$$G_t(x) = \{ r : u_t(x, r) \le C/\lambda_t \}.$$

The global KDE uses uniform weights over a pooled residual bank, with at most 1024 components.

Zonotopes. Retrieved residuals define a weighted center $c_t(x)$ and a square PCA generator matrix $A_t(x)$. The base zonotope is

$$Z_t(x) = c_t(x) + A_t(x)[-1, 1]^6.$$

We score the multiplicative excess outside this base geometry by

$$u_t(x, r_t) = \left(\left\| A_t(x)^{-1}(r_t - c_t(x)) \right\|_\infty - 1 \right)_+.$$

SNSA calibration therefore produces

$$A_t(x) = c_t(x) + \left(1 + \frac{C}{\lambda_t}\right) A_t(x)[-1, 1]^6.$$

We use ridge 10^{-6} when constructing the generators.

Input-space localization baselines. To isolate the learned retrieval representation from the downstream geometry, we also construct ellipsoids by retrieving neighbors directly from the flattened 806-dimensional nominal state and control trajectory. We evaluate raw Euclidean distance and Euclidean distance after standardizing each state and control channel using geometry-bank statistics. All residual geometry construction and conformal calibration are otherwise unchanged. Raw and standardized input-space retrieval obtain mean log volumes -26.66 compared with -25.82 for global geometry and -33.25 for learned retrieval. Their held-out simultaneous coverages are 98.7%, 98.4%, and 99.1%, respectively.

Retrieval objective ablation. To isolate the effect of the representation-learning objective, we compare energy-score training against two Auto Encoder (AE) baselines. The Covariate AE learns a representation by reconstructing the input covariate X, while the Residual AE maps X to its corresponding residual R and minimizes residual reconstruction error. Both baselines use the encoder's latent representation for retrieval.

All models have matched encoder parameter counts, an 8-dimensional latent space, identical training episodes and settings, and 250 training epochs. Retrieval temperatures are tuned independently. The autoencoder baselines have approximately twice the total parameters due to their decoders. Retrieval, ellipsoidal geometry construction, and conformal calibration are otherwise identical.

Table 6 shows that energy-score training achieves 5.64–7.43 lower mean log-volume across the 90%, 95%, and 99% coverage targets, with coverage near nominal. At 95%, energy achieves -34.899, compared with -28.903 for Covariate AE and -27.466 for Residual AE. This supports learning representations directly for conditional residual-law retrieval, rather than input reconstruction or pointwise residual prediction.

Table 6: Retrieval objective ablation for quadcopter ellipsoids. Mean log-volume at each target coverage; empirical simultaneous coverage (%) is shown in parentheses.

Target	Energy	Covariate AE	Residual AE
90%	$\mathbf{-35.918}$ (91.8%)	-29.795 (90.2%)	-28.527 (90.3%)
95%	$\mathbf{-34.899}$ (95.7%)	-28.903 (94.8%)	-27.466 (95.5%)
99%	$\mathbf{-32.522}$ (99.0%)	-26.887 (98.6%)	-25.560 (99.3%)

Capacity sweeps. We compare FacCP against resCONTRA (Fang et al., 2025) and Covariance Prediction CP (Srinivasan et al., 2026; Messoudi et al., 2022). resCONTRA learns a query-dependent flow transformation. For resCONTRA, we estimate the volume of the flow-defined conformal region by quasi-Monte Carlo integration of the inverse-flow Jacobian over the conformal latent region. Covariance Prediction CP directly predicts a query-dependent covariance parameterization and applies a Mahalanobis score. At each sweep point we choose architectures with closely matched learned parameter counts and report means over 25 evaluation repeats. The target trajectory coverage is 99%. Table 7 reports all completed capacity-sweep points.

Data sweeps. We fix capacity near the equal-volume crossover of each capacity sweep. For resCONTRA the matched models have approximately 2.03×10^6 parameters, while the Covariance Prediction CP comparison uses approximately 2.03×10^5. We vary the total data budget while holding validation, calibration, and test data fixed. Each FacCP budget includes a fixed 2,000-trajectory geometry bank, with the remaining trajectories used to train the retrieval representation. The baseline uses the corresponding total budget for direct model training. The data sweeps use 97.5% target trajectory coverage and 50% of the calibration trajectories for SNSA normalization. Results are averaged over 25 evaluation repeats.

α-local pivotality. At 99% target coverage and for a calculated $\eta_n = 0.0341$, we define query difficulty as the sum of the root-mean-square standardized nominal-state and control magnitudes over the query horizon. Queries are grouped into ten difficulty bins using decile boundaries fitted on the reference geometry split. Over these difficulty bins, $\widehat{\Delta}_{\alpha,\eta_n}$ is 0.0942 for FacCP and 0.1570 for Global; the corresponding $\widehat{\Delta}_{\alpha,\eta_n} + \eta_n$ values are 0.1283 and 0.1911. At the fitted threshold, pooled coverage is $0.990/0.990$ (FacCP/Global), with maximum difficulty-bin coverage gaps $0.046/0.068$. These diagnostics instantiate the empirical analogue of Cor. 7 above.

C.3.1 3D Quadcopter Experiment Details

Setup. The 3D quadcopter has a 12-dimensional state and 4-dimensional control with horizon $H = 200$. The nominal rollout and control sequence give

$$201 \cdot 12 + 200 \cdot 4 = 3212$$

query coordinates, while the residual trajectory contains $200 \cdot 12 = 2400$ coordinates. At each time step we construct a 12-dimensional ellipsoid and apply the same trajectory-level conformal cali-

Table 7: Planar quadcopter model-capacity sweeps. FacCP is compared against resCONTRA and Covariance Prediction CP at approximately matched parameter counts. $\log V$ denotes mean 6-dimensional log volume (lower is better); coverage denotes simultaneous trajectory coverage in percent (target: 99%).

Parameters (N_θ)		$\log V$		Coverage (%)	
FacCP	Baseline	FacCP	Baseline	FacCP	Baseline
FacCP vs. resCONTRA					
101	100	-25.85	-17.96	98.9	98.9
296	298	-28.37	-17.19	99.0	98.9
1,006	1,009	-30.83	-16.53	98.9	99.0
1,990	1,986	-32.11	-20.70	98.6	99.0
5,002	4,995	-32.80	-24.07	98.7	98.9
13,040	12,952	-32.78	-25.75	98.9	98.9
34,528	34,484	-33.11	-27.28	98.7	99.3
66,256	66,132	-32.84	-27.55	98.6	99.1
169,096	169,044	-33.01	-28.34	98.8	99.0
262,568	262,756	-33.05	-28.30	98.6	99.1
376,520	375,688	-32.77	-29.23	98.4	99.0
1,037,128	1,038,100	-32.65	-31.12	99.1	98.9
2,025,416	2,025,284	-32.93	-33.10	98.7	99.1
4,122,248	4,122,408	-32.56	-32.91	98.8	99.1
FacCP vs. Covariance Prediction CP					
2,056	2,363	-32.39	-26.86	98.6	99.0
7,176	7,771	-32.48	-27.95	99.0	98.8
26,632	27,803	-32.95	-28.90	98.6	98.9
102,408	104,731	-33.35	-29.55	98.5	99.2
201,992	204,315	-33.09	-31.68	98.6	99.0
450,440	453,915	-32.75	-32.23	98.9	98.9
797,192	801,819	-32.75	-33.47	99.0	99.1
1,785,608	1,792,539	-32.36	-33.43	98.7	99.0
3,167,240	3,176,475	-32.62	-33.85	99.1	99.2
7,110,152	7,123,995	-32.24	-33.37	98.9	99.0

bration as in Appendix C.2, so a single conformal event controls the complete 200-step trajectory. We use at most 5,000 geometry trajectories, 2,000 calibration trajectories, and 1,000 test trajectories, with 35% of calibration trajectories used for time normalization and 99% target simultaneous coverage. Reported $\log V$ is the mean per-time-step 12-dimensional log volume.

Capacity sweeps. Table 9 reports results at approximately matched parameter counts. FacCP substantially outperforms Mahalanobis CP throughout the tested range. At approximately 2.5×10^3 parameters, mean log volumes are -74.50 and -62.15, respectively. Although Mahalanobis improves with capacity, its best observed value is -76.09 at approximately 6.4×10^5 parameters, compared with -77.96 for FacCP. Even at 1.6×10^6 parameters, the gap persists (-78.19 vs. -74.67). resCONTRA exhibits a different trend. At 4.3×10^3 parameters, its mean log volume is -32.69 compared with -76.02 for FacCP. The gap narrows substantially by 2.7×10^4 parameters (-71.97 vs. -78.29) and nearly closes around 6×10^4 (-78.14 vs. -78.71). At higher capacities, resCONTRA surpasses FacCP, reaching -85.18 at 2.2×10^5 parameters while FacCP remains near -78. These results suggest that FacCP is more parameter-efficient at low capacity, whereas sufficiently expressive flow models can ultimately achieve tighter sets.

Data sweeps. We fix model capacity at approximately 1.6×10^6 parameters for Mahalanobis CP and 6×10^4 for resCONTRA, varying the total training-data budget while keeping validation, calibration, and test sets fixed. FacCP consistently achieves smaller volumes than Mahalanobis CP, both at 20% of the data (-73.16 vs. -67.40) and at the full budget (-78.02 vs. -74.67). Against resCONTRA, FacCP is more efficient with limited data (-70.58 vs. -67.74 at 20%), but the gap narrows as the training budget increases. The methods are approximately matched near 70%, beyond

Table 8: Planar quadcopter training-data sweeps near the model-capacity crossover. FacCP is compared against resCONTRA and Covariance Prediction CP at varying data budgets. Training counts exclude FacCP's fixed retrieval bank of 2,000 trajectories. $\log V$ denotes mean 6-dimensional log volume (lower is better); coverage denotes simultaneous trajectory coverage in percent.

	Training trajectories		$\log V$		Coverage (%)	
Budget	FacCP	Baseline	FacCP	Baseline	FacCP	Baseline
FacCP vs. resCONTRA						
20%	700	2,700	-32.46	-27.11	97.1	97.3
30%	2,050	4,050	-32.04	-27.24	97.1	97.7
40%	3,400	5,400	-33.01	-29.21	97.2	97.8
50%	4,750	6,750	-32.70	-29.95	97.5	97.8
60%	6,100	8,100	-32.86	-30.40	97.6	97.9
70%	7,450	9,450	-32.54	-31.76	97.4	98.1
80%	8,800	10,800	-32.39	-32.34	97.9	97.9
90%	10,150	12,150	-32.44	-33.27	97.6	97.9
100%	11,500	13,500	-32.40	-33.40	97.3	97.9
FacCP vs. Covariance Prediction CP						
20%	700	2,700	-32.84	-29.02	97.0	97.5
30%	2,050	4,050	-31.37	-29.67	97.2	97.4
40%	3,400	5,400	-32.92	-30.02	97.1	97.5
50%	4,750	6,750	-32.99	-30.21	97.2	97.5
60%	6,100	8,100	-32.85	-30.72	97.2	97.8
70%	7,450	9,450	-32.77	-31.24	97.3	97.3
80%	8,800	10,800	-32.91	-31.75	97.3	97.0
90%	10,150	12,150	-32.86	-31.83	97.4	97.7
100%	11,500	13,500	-32.73	-32.12	97.4	97.8

Table 9: 3D quadcopter model-capacity sweeps. FacCP is compared against Mahalanobis CP and resCONTRA at approximately matched parameter counts. $\log V$ denotes mean per-time-step 12-dimensional log volume (lower is better); coverage denotes simultaneous coverage of the complete 200-step trajectory in percent (target: 99%).

Parameters (N_θ)		$\log V$		Coverage (%)	
FacCP	Baseline	FacCP	Baseline	FacCP	Baseline
FacCP vs. Covariance Prediction CP					
2,514	2,506	-74.50	-62.15	99.1	98.9
69,972	69,961	-77.74	-72.94	99.2	99.4
100,002	100,004	-77.39	-71.43	99.1	99.0
160,008	159,974	-77.83	-74.36	99.2	99.1
250,013	249,982	-78.39	-74.45	99.2	99.2
400,006	399,991	-77.53	-74.95	99.1	99.0
639,997	639,994	-77.96	-76.09	99.1	99.3
1,000,006	1,000,004	-78.45	-76.03	99.0	99.2
1,600,023	1,599,991	-78.19	-74.67	99.1	98.9
FacCP vs. resCONTRA					
4,287	4,280	-76.02	-32.69	99.1	99.1
26,714	26,744	-78.29	-71.97	99.0	99.1
40,152	40,155	-77.16	-72.07	99.0	99.1
60,022	60,064	-78.71	-78.14	99.2	99.1
89,964	89,978	-77.89	-81.44	99.2	99.1
140,004	139,992	-77.89	-83.58	99.1	99.3
220,017	219,984	-77.88	-85.18	99.0	98.9

which resCONTRA achieves slightly smaller volumes. Across both sweeps, simultaneous trajectory coverage remains near the 99% target.

Table 10: 3D quadcopter training-data sweeps at fixed model capacity. FacCP is compared against Mahalanobis CP ($\sim$ 1.6M parameters) and resCONTRA ($\sim$ 60k parameters) at varying data budgets. $\log V$ denotes mean per-time-step 12-dimensional log volume (lower is better); coverage denotes simultaneous coverage of the complete 200-step trajectory in percent (target: 99%).

Training data		$\log V$		Coverage (%)	
Budget (%)	Total episodes	FacCP	Baseline	FacCP	Baseline
FacCP vs. Covariance Prediction CP					
20	5,400	-73.16	-67.40	99.3	99.1
30	8,100	-73.87	-70.51	99.2	99.1
40	10,800	-77.00	-70.13	99.2	99.2
50	13,500	-77.48	-71.68	99.2	98.8
60	16,200	-77.58	-72.65	99.1	99.0
70	18,900	-77.91	-73.34	99.2	99.1
80	21,600	-77.22	-74.19	99.2	99.0
90	24,300	-77.38	-74.86	99.2	99.4
100	27,000	-78.02	-74.67	99.1	98.9
FacCP vs. resCONTRA					
20	5,400	-70.58	-67.74	99.1	99.2
30	8,100	-72.27	-71.61	99.0	99.1
40	10,800	-76.10	-72.13	99.1	99.2
50	13,500	-76.08	-73.71	99.0	99.1
60	16,200	-76.66	-75.79	99.1	99.1
70	18,900	-76.63	-76.86	99.1	99.1
80	21,600	-76.76	-78.24	99.1	99.2
90	24,300	-77.02	-77.51	99.1	99.1
100	27,000	-77.86	-78.39	99.1	99.1

α-local pivotality. At 99% target coverage, $\eta_n = 0.0280$. Over ten reference-fit difficulty bins, $\widehat{\Delta}_{\alpha,\eta_n}$ is 0.1099 for FacCP and 0.3166 for Global; the corresponding $\widehat{\Delta}_{\alpha,\eta_n} + \eta_n$ values are 0.1378 and 0.3446. At the fitted threshold, pooled coverage is $0.994/0.989$ (FacCP/Global), with maximum difficulty-bin coverage gaps $0.041/0.085$. These diagnostics instantiate the empirical analogue of Cor. 7 above.

D AUTONOMOUS-DRIVING SUPPLEMENTARY RESULTS

The main comparison uses six candidates and fixed pricing. Here we give the construction behind those results, separate pricing from localization, and examine where the gains do and do not persist. The main comparison includes pricing for Global and Local. Here we show the corresponding unpriced ablation. MoE-CP retains its fixed $\eta = 1$ setting even where this parameter is omitted from plot labels.

D.1 DATASET AND ENCODER

The processed data contains 183,333 training and 22,979 validation scenarios. We train FacCP encoder on the training split. We partition the validation split into 8,042 scenes reserved for encoder validation split and checkpoint selection, 6,894 reference scenes, 4,596 calibration scenes, and 3,447 internal test scenes. We have 900 scenes for conformal calibration where dev set uses separate subsets of the calibration pool and 1,000 scenes for method evaluation. All these partitions are disjoint at the scenario level.

The encoder takes focal history 21×18, neighbor histories $32 \times 21 \times 18$, map features $128 \times 20 \times 9$ and an anchor 60×2, with validity masks and map types. The 7.52M-parameter network uses width-256 tokens, four scene-attention layers and four trajectory-decoder layers, followed by a unit-normalized 16-dimensional embedding. We first pretrain the encoder for 15 epochs using masked reconstruction. For each scene, we retain 25% of the valid agent-history and map tokens as input and train the model to reconstruct the continuous features of the remaining masked tokens.

Reconstruction uses a temporary two-layer decoder with hidden width 128, which is discarded after pretraining. After this, 10 epochs of energy fine-tuning follow. The checkpoint is fixed across all targets.

For whitened residual trajectories $u_i \in \mathbb{R}^{120}$, fine-tuning uses

$$\mathcal{L}_i = \sum_j a_{ij} \frac{\|u_i - u_j\|_2}{\sqrt{120}} - \frac{1}{2} \sum_{j,\ell} a_{ij} a_{i\ell} \frac{\|u_j - u_\ell\|_2}{\sqrt{120}}.$$

Here, a_{ij} is the normalized weight assigned to reference scene j based on its embedding distance from query scene i. Each minibatch is split into two disjoint halves, with each half providing reference scenes for queries in the other. The reconstruction decoder is used only during pretraining and is then discarded. The subsequent objective trains the embeddings for residual-based retrieval; it does not directly minimize the footprint of the final conformal tubes.

D.2 Reference errors, geometry and SNSA

Global SNSA uses equally weighted reference scenes to estimate a covariance and a 95th-percentile Mahalanobis residual scale at each timestep. Local SNSA uses learned embedding weights to estimate both quantities separately for each query candidate. Global and Local SNSA use the same predicted paths as tube centers. They differ in how they weight reference errors to estimate each tube's shape and scale. To describe this construction, let $\mu_{j,k,t}$ denote the predicted position at time t for candidate k in reference scene j. Each scene contributes one residual path and one embedding:

$$e_{j,t} = y_{j,t} - \mu_{j,\kappa_j,t}, \qquad \bar{z}_j = h_\theta(x_j, \kappa_j), \qquad \kappa_j = \arg\min_k \frac{1}{60} \sum_t \|y_{j,t} - \mu_{j,k,t}\|_2.$$

The minimizing mode is fixed across the whole path. For the single-mode comparison, replace this choice with $\arg\max_k \pi_{j,k}$ and predict only the top-probability query mode. Neither procedure uses a test label when constructing its set. Query embeddings are $z_k(x) = h_\theta(x, k)$ where h_θ is our FacCP encoder.

Covariance and weights. For a query mode k, Local assigns weights to all N bank scenes:

$$w_{k,j}(x) = \frac{\exp(-\|z_k(x) - \bar{z}_j\|_2^2/\tau)}{\sum_{\ell=1}^N \exp(-\|z_k(x) - \bar{z}_\ell\|_2^2/\tau)}, \qquad \tau = 1.0 \tag{27}$$

The same weights apply at all times, without top-k truncation or a uniform mixture. Write $\bar{e}_t = N^{-1} \sum_j e_{j,t}$ and $m_{k,t}(x) = \sum_j w_{k,j}(x) e_{j,t}$. The covariance estimates are

$$C_t^G = \frac{\sum_j (e_{j,t} - \bar{e}_t)(e_{j,t} - \bar{e}_t)^\top}{N - 1} + \varepsilon I_2, \tag{28}$$

$$C_{k,t}^L(x) = \frac{\sum_j w_{k,j}(x)(e_{j,t} - m_{k,t})(e_{j,t} - m_{k,t})^\top}{1 - \sum_j w_{k,j}(x)^2} + \varepsilon I_2, \qquad \varepsilon = .01\,\mathrm{m}^2. \tag{29}$$

Global estimates one covariance matrix for each of the 60 timesteps and uses these same matrices for every scene and candidate path. Local instead estimates a separate sequence of covariance matrices for each candidate path in each query scene. In both methods, reference errors are centered only to estimate covariance; the predicted MTR paths remain unchanged. Local uses the weighted covariance directly, without blending it with the Global estimate.

SNSA: normalizing residual scale over time. For either geometry, the query radius is $\rho_{k,t}(x, y) = \sqrt{(y_t - \mu_{k,t})^\top C_{k,t}(x)^{-1}(y_t - \mu_{k,t})}$. SNSA estimates a separate residual scale at each time:

$$r_{j,t}^G = \sqrt{e_{j,t}^\top (C_t^G)^{-1} e_{j,t}}, \qquad s_t^G = Q_{.95}(r_{1,t}^G, \ldots, r_{N,t}^G) + 10^{-6}, \tag{30}$$

$$r_{j,t}^L = \sqrt{e_{j,t}^\top (C_{j,t}^{L,-j})^{-1} e_{j,t}}, \qquad s_{k,t}^L(x) = Q_{.95}^{w_k(x)}(r_{1,t}^L, \ldots, r_{N,t}^L) + 10^{-6}. \tag{31}$$

Here $C_{j,t}^{L,-j}$ is bank scene j's local covariance after excluding its own residual and renormalizing the weights. Thus stored local radii use the reference scenes' own geometries; the query weights then select their weighted percentile. Global scales are shared across query scenes and modes; Local scales depend on the query mode. Q^w uses weighted interpolation, defined in the interpolation paragraph below. The normalizer quantile .95 estimates scale and consequently SNSA uses $\lambda = 1/s$.

Interpolation and bank draws. For weighted type-7 interpolation, sort radii $z_1 \le \cdots \le z_N$ with normalized positive weights ω_j, set $c_j = \sum_{\ell<j} \omega_\ell$ and $d = 1-\omega_N$, and let $h = \min\{N-1, \max\{j : c_j \le pd\}\}$. Then

$$Q_p^w = z_h + \mathrm{clip}_{[0,1]}\left(\frac{pd - c_h}{\omega_h}\right)(z_{h+1} - z_h).$$

Uniform weights recover ordinary type-7 interpolation. Shrinking every scale by the same factor does not shrink the calibrated sets without pricing: at fixed geometry, $s \mapsto cs$ gives $\widehat{q} \mapsto \widehat{q}/c$. Efficiency therefore depends on relative scale allocation and covariance shape, not uniformly smaller scales.

For each paired draw, methods retain the same random 90% of bank scenes; the omitted 10% is unused.

D.3 Calibration and pricing

With 900 calibration scenes and 60 timesteps, a Bonferroni correction yields unbounded finite-rank thresholds at targets $\ge 95\%$. Even when finite, simultaneous per-time coverage allows the covering candidate to change over time. Pricing adjusts each candidate's tube budget according to its predicted probability. We calibrate the resulting scene score on the separate calibration set. The score below covers both the priced results in the main table and the unpriced control, obtained by setting $\gamma = 0$.

Whole-path calibration. With $s_{k,t} = s_t^G$ or $s_{k,t}^L(x)$, $n = 900$ equally weighted calibration scenes give the score, threshold and prediction set, with pricing fee $b_k(x) = -\gamma \log \pi_k(x)$:

$$S_\gamma(x,y) = \min_{k\le 6}\left[\max_{t\le 60}\frac{\rho_{k,t}(x,y)}{s_{k,t}(x)} + b_k(x)\right], \qquad \widehat{q}_{\delta,\gamma} = S^{\mathrm{cal}}_{\gamma,(\lceil (n+1)(1-\delta)\rceil)}, \tag{32}$$

$$\widehat{C}_{\delta,\gamma}(x) = \bigcup_{k:\, b_k(x)\le \widehat{q}_{\delta,\gamma}} \{y : \rho_{k,t}(x,y) \le (\widehat{q}_{\delta,\gamma} - b_k(x)) s_{k,t}(x)\ \forall t\}. \tag{33}$$

If the required calibration rank exceeds n, the threshold is $+\infty$. We compare the unpriced score ($\gamma = 0$) with fixed pricing ($\gamma = .10$), calibrating a separate threshold for each while keeping the encoder, covariance matrices and SNSA scales fixed. The scales determine the relative tube radii, while $\widehat{q}_{\delta,\gamma}$ sets their overall size to provide marginal whole-path coverage under exchangeability, with the scoring rule fixed before calibration. This guarantee applies across scenes, rather than separately within each scene group. Test labels are used only to evaluate coverage, not to construct the prediction sets.

Only the score and calibrated threshold change between $\gamma = 0$ and $\gamma = .10$. The same geometry, normalizers, candidates and calibration scenes are retained.

Why calibrate the complete path? Per-time coverage does not imply simultaneous coverage. A Bonferroni construction assigns miscoverage $\delta/60$ to each timestep and uses calibration rank $\lceil (n+1)(1-\delta/60)\rceil$. With $n = 900$, this rank exceeds 900 at the 95%, 97% and 99% targets, requiring an infinite threshold under the finite-rank convention. At 90% the rank is 900. Moreover, simultaneous membership in per-time unions still permits mode switching; it does not imply membership in one complete candidate tube. Our maximum-over-time, minimum-over-modes score calibrates the desired event directly.

D.4 What do the representation and training objective contribute?

We first ask whether the learned embeddings improve on features already available from pretraining or motion forecasting. We then ask whether energy fine-tuning helps compared with another way of learning which reference errors to retrieve. Both comparisons use the full-data encoder: 15 epochs of masked reconstruction pretraining followed by 10 epochs of fine-tuning on the training scenes.

Comparison setup. We keep the six MTR paths, Local SNSA construction, normalizer quantile and pricing strength fixed. All learned representations use retrieval temperature $\tau = 1.0$, and all methods, including Global SNSA, use pricing $\gamma = .10$. Evaluation uses the same test scenes, reference scenes and calibration scenes. Each entry below is the percentage reduction in mean time-integrated footprint relative to Global SNSA, calculated within each of 16 paired reference-bank subsets and then averaged where positive values mean smaller footprints. No model or temperature is selected from test set.

Pretrained representations. The MAE baseline uses the same pretrained checkpoint from which FacCP Stage 1 encoder is fine-tuned, before any energy training. We flatten its 161×256 scene features into a 41,216-dimensional vector, zeroing invalid tokens. The MTR baseline uses the frozen forecaster's 256-dimensional focal-agent feature. Both vectors are unit-normalized and used directly for retrieval, without a learned projection. MAE and MTR provide one embedding per scene; Complete FacCP produces a 16-dimensional embedding for each scene–candidate pair.

Representation	Dimension	95%	97%	99%
Frozen MTR	256	3.74 / 95.20	10.85 / 96.50	9.80 / 99.12
FacCP Stage 1 (Pretrained MAE)	41,216	-0.12 / 95.25	2.18 / 97.01	5.32 / 99.12
Proposed FacCP (MAE + energy)	16	**14.15** / 94.87	**12.92** / 96.92	**27.50** / 98.85

Table 11: Entries are footprint reductions (%) relative to Global SNSA with empirical whole-path coverage on test scenes.

Table 11 shows larger reductions with energy-trained FacCP at all three targets. The learned embedding with the fine-tuning objective therefore gives more efficient tubes in this comparison.

Training objective. To isolate the objective, we fine-tune a the same FacCP Stage 1 encoder using a residual-neighbor KL loss. For each training query, distances between its whitened residual trajectory and reference residuals are converted into target neighbor probabilities. KL training makes the embedding-based retrieval weights match those probabilities. For query i, we convert residual distances into target retrieval probabilities and train the embedding weights to match them:

$$p_{ij} = \frac{\exp(-d_{ij}/T)}{\sum_{\ell} \exp(-d_{i\ell}/T)} \qquad T = 0.5, \qquad \mathcal{L}_{\mathrm{KL}} = \frac{1}{|\mathcal{Q}|} \sum_{i \in \mathcal{Q}} \sum_{j} p_{ij} \log \frac{p_{ij}}{w_{ij}}.$$

Here $d_{ij} = \|u_i - u_j\|_2 / \sqrt{120}$ compares whitened 60-step residual trajectories, and $w_{ij} \propto \exp(-\|h_\theta(x_i, k_i) - h_\theta(x_j, k_j)\|_2^2 / \tau)$ are normalized embedding-based retrieval weights. $\mathcal{Q}$ is the set of training queries in one half of a minibatch and both j and ℓ index reference examples in the other half of the minibatch. p_{ij} are fixed during backpropagation and gradients update h_θ through w_{ij}. Energy training instead scores the weighted residual distribution directly, using the objective in Appendix D.1. The two runs share the architecture, training scenes, minibatch order, optimizer settings and ten-epoch budget. Their encoders are then frozen, and the same Local SNSA and calibration procedures are applied.

Method	95%	97%	99%
FacCP Ablation (MAE + KL loss)	0.76 / 95.02	-0.61 / 97.05	8.86 / 99.16
Proposed FacCP (MAE + energy)	**14.15** / 94.87	**12.92** / 96.92	**27.50** / 98.85

Table 12: Entries are footprint reductions (%) relative to Global SNSA at nominal target coverage on test. Negative values indicate larger footprints than Global.

Our proposed FacCP with energy fine-tuning gives most reductions (Table 12). The result suggests that training retrieval to predict the residual distribution is more useful here than training it to imitate a prescribed neighbor-weight pattern.

D.5 SENSITIVITY TO TRAINING DATA AND MODEL CAPACITY

We examine how FacCP changes with the amount of training data and the capacity of its representation model. These experiments vary one component at a time while keeping retrieval, geometry construction and calibration fixed.

Experimental setup. For the data sweep, we train the full architecture on nested subsets containing 10% to 100% of the 183,333 available training scenes. Both pretraining and fine-tuning use only the selected subset. Feature normalization is also performed using that subset. For the capacity sweep, we use all training scenes and vary the hidden width of the scene encoder and candidate decoder, targeting 10% to 100% of the full model's parameter count. The number of layers and attention heads remains fixed, and every model produces a 16-dimensional embedding. The masked-reconstruction decoder retains width 128. Since hidden widths must be divisible by eight, actual parameter fractions differ slightly from the requested fractions.

Every configuration receives 15 epochs of masked reconstruction pretraining and ten epochs of energy fine-tuning. All runs use one fixed training seed. We retain the same six frozen MTR predictions and use Local SNSA with retrieval temperature $\tau = 1.0$, pricing strength $\gamma = .10$, and a 95th-percentile normalization scale.

Reported quantities. Each table entry reports mean footprint in m^2 s, percentage footprint reduction relative to Global SNSA, and empirical whole-path coverage (%), in that order. The reduction is

$$100\left(1 - \frac{\overline{V}_{\text{FacCP}}}{\overline{V}_{\text{Global}}}\right),$$

so positive values indicate smaller footprints. Global SNSA provides a fixed reference for expressing the scale of the changes across FacCP configurations.

Fraction (%)	Scenes	Parameters	95%	97%	99%
10	18,333	7,524,624	610.9 / 10.11 / 94.92	819.0 / 6.92 / 96.98	1,510.8 / 9.79 / 99.25
20	36,667	7,524,624	591.1 / 13.02 / 94.95	778.2 / 11.55 / 96.87	1,363.6 / 18.58 / 99.19
30	55,000	7,524,624	587.5 / 13.55 / 94.57	787.2 / 10.53 / 96.92	1,328.7 / 20.66 / 99.07
40	73,333	7,524,624	583.4 / 14.16 / 94.72	751.3 / 14.61 / 96.87	1,338.0 / 20.11 / 99.10
50	91,666	7,524,624	601.3 / 11.52 / 95.24	753.0 / 14.41 / 96.87	1,389.4 / 17.04 / 99.19
60	110,000	7,524,624	581.0 / 14.51 / 94.89	748.0 / 14.99 / 96.87	1,285.8 / 23.23 / 99.01
70	128,333	7,524,624	602.1 / 11.40 / 95.21	761.4 / 13.46 / 96.72	1,213.5 / 27.54 / 98.96
80	146,666	7,524,624	556.4 / 18.13 / 94.40	761.4 / 13.45 / 96.95	1,270.2 / 24.16 / 98.98
90	165,000	7,524,624	578.4 / 14.89 / 94.60	748.7 / 14.90 / 96.92	1,159.4 / 30.77 / 98.78
100	183,333	7,524,624	587.1 / 13.61 / 94.95	784.8 / 10.80 / 97.27	1,379.2 / 17.65 / 99.16

Table 13: Training-data fraction (full architecture). Cells give mean footprint (m^2 s), percentage reduction versus Global SNSA, and empirical whole-path coverage (%)

Effect of training data. Table 13 shows that FacCP already reduces footprint relative to Global SNSA when trained on only 10% of the available scenes. Increasing the training fraction improves on this starting point, but the gains are not monotonic: using all training scenes does not produce the smallest footprint at any target.

Effect of model capacity. Table 14 shows that increasing model capacity does not consistently reduce footprint. At the 95% target, the smallest model achieves nearly the same footprint as the full model, despite using roughly one tenth of its parameters. All configurations retain a 16-dimensional retrieval embedding, so these differences concern the capacity used to learn the representation, not its output dimension. The results suggest that a compact encoder can capture useful information for localization; they do not establish that a larger encoder improves the resulting sets under the same training recipe.

What these sweeps establish. These experiments quantify the sensitivity of the final calibrated sets to training data and model capacity under a fixed procedure.

Fraction (%)	Scenes	Parameters	95%	97%	99%
10	183,333	746,336	588.6 / 13.39 / 94.60	774.5 / 11.97 / 97.10	1,221.1 / 27.09 / 98.93
20	183,333	1,453,440	605.3 / 10.93 / 95.13	767.0 / 12.82 / 96.95	1,263.7 / 24.54 / 99.04
30	183,333	2,136,984	580.2 / 14.63 / 94.72	729.2 / 17.12 / 96.69	1,197.7 / 28.49 / 98.90
40	183,333	2,951,856	589.1 / 13.31 / 94.87	792.4 / 9.93 / 97.24	1,133.1 / 32.34 / 98.78
50	183,333	3,898,056	592.8 / 12.77 / 94.95	791.1 / 10.08 / 97.30	1,246.8 / 25.55 / 99.01
60	183,333	4,601,816	588.3 / 13.44 / 94.66	796.2 / 9.51 / 97.21	1,220.1 / 27.15 / 98.96
70	183,333	5,363,944	586.9 / 13.63 / 94.81	757.8 / 13.87 / 96.87	1,256.5 / 24.98 / 98.98
80	183,333	6,184,440	587.0 / 13.63 / 94.87	753.3 / 14.38 / 97.07	1,207.4 / 27.90 / 98.98
90	183,333	6,616,576	604.8 / 11.00 / 95.01	748.3 / 14.95 / 96.95	1,203.7 / 28.13 / 98.78
100	183,333	7,524,624	587.1 / 13.61 / 94.95	784.8 / 10.80 / 97.27	1,379.2 / 17.65 / 99.16

Table 14: Model size (all training data). Cells give mean footprint (m^2 s), percentage reduction versus Global SNSA, and empirical whole-path coverage (%).

D.6 Baselines and measurement

FORCE-OPT (Chakraborty et al., 2026) is implemented through its offline allocation and per-time calibration core, without its online Bayesian monitor or planning system. Its whole-path coverage here is an additional evaluation event. MoE-CP (Kong et al., 2026) uses randomized gating-similarity calibration on the unpriced Global-SNSA score, with MTR probabilities as the gate. We retain the query infinity atom required by that construction. The reported $\eta = 1$ setting is exploratory: larger settings produced unbounded sets when weighted calibration support was insufficient. In our MoE-CP implementation, MTR candidate probabilities determine calibration-scene weights, while covariance and normalization scale remain shared across queries at each timestep.

The single-mode comparisons ask a further question: how do timestep tubes compare with sets that constrain an entire trajectory jointly? NLE (Messoudi et al., 2022) fits one joint 120D error ellipsoid using 5% nearest neighbors in standardized anchor/context features, blending 95% local with 5% pooled covariance; it retains empirical interpolated calibration. MVCS (Braun et al., 2026) learns a joint transform with frozen centers, bank whitening and finite-rank calibration.

We measure footprint as the spatial area occupied by the prediction set, integrated over time: $V(x) = 0.1 \sum_t \mathrm{area}(\bigcup_k E_{k,t})$, in m^2 s. Here, $E_{k,t}$ denotes the ellipse for candidate k at timestep t, and overlapping regions are counted only once. For NLE and MVCS, footprint uses the two-dimensional projections of the joint trajectory ellipsoid at each timestep, whereas coverage requires membership in the full joint set. Rank-6 coverage is evaluated on the test scenes where the least-probable MTR candidate has the lowest average displacement error relative to the observed future. The single-mode evaluation uses this same scene subset, while predicting only the highest-probability path.

We only present NLE and MVCS for single mode since that gives all methods one predicted path to describe uncertainty around. In our NLE and MVCS-L2 implementations construct a joint ellipsoid around the top-probability trajectory., which makes single mode a direct comparison of different uncertainty constructions around the same forecast. With six modes, the forecast might include turn left, continue straight, and turn right. One ellipsoid spanning those alternatives would inflate those volumes by including substantial space between them, where no candidate path goes.

D.7 Footprint and coverage across operating points

We compare Global SNSA and FacCP on footprint and target versus empirical coverage (Figure 6). The left panel plots footprint against empirical coverage. At the 95%, 97% and 99% targets, FacCP reduces footprint by 13.7%, 13.7% and 29.2%, respectively, relative to Global SNSA. The right panel shows differences between target and empirical coverage.

D.8 What computational cost does FacCP add?

Constructing tubes for a new scene. We measure the additional conformal computation after the frozen MTR paths and probabilities are available. For FacCP, this includes processing the raw scene tensors through h_θ, producing six candidate embeddings, comparing them with the reference bank,

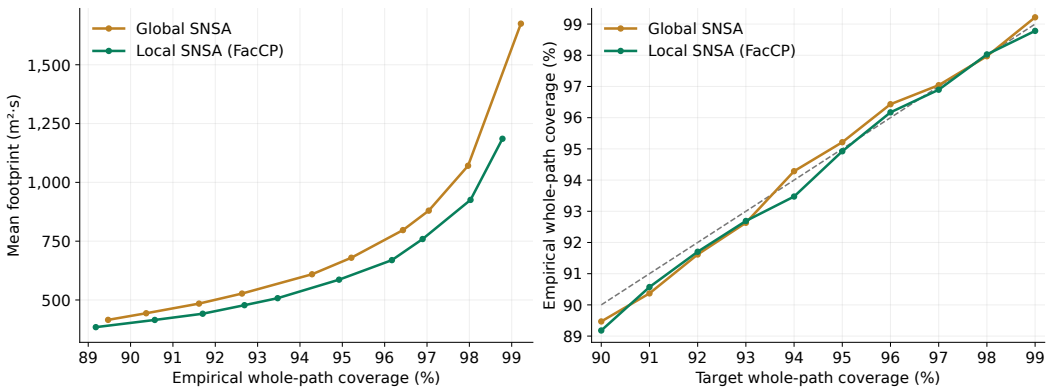


Figure 6: Global SNSA and Local SNSA (FacCP) on Argoverse 2. Left: mean footprint versus empirical whole-path coverage, varying the threshold while keeping each method's geometry fixed. Right: empirical coverage at targets from 90% to 99%.

estimating covariances and SNSA scales, and constructing all six 60-step tubes. The calibrated threshold is fixed before prediction.

We preload the scene inputs, reference bank and calibration data onto one RTX 4090. After 32 warmup scenes, we time each of the test scenes individually, synchronizing the GPU before and after each prediction. FacCP adds 3.581 ms per scene on average relative to Global SNSA.

Method	Mean (ms)	P95 (ms)	Additional neural calls
Global SNSA	0.034	0.037	None
Local SNSA (FacCP)	3.615	3.729	1 scene + 1 batched candidate call

Table 15: Per-scene latency for constructing all six 60-step tubes on an RTX 4090, averaged over all test scenes

Training. The archived optimization timers for the selected model record 18.13 minutes for 15 epochs of MAE pretraining and 22.27 minutes for ten epochs of energy fine-tuning on 183,333 scenes, totaling 40.40 minutes on a RTX 4090. Global SNSA requires no additional neural-network training; it only estimates covariance and normalization scales from the reference bank.

D.9 Does localization balance coverage across scenes?

A smaller average footprint does not by itself indicate more uniform coverage across scene types. We therefore examine both the similarity of score distributions near the target quantile and the empirical coverage within each scene bin.

Let $\widehat{F}_g$ denote the empirical score CDF in bin g, and $\widehat{F}$ the corresponding CDF over the full TEST cohort. We measure their largest discrepancy near the requested coverage level:

$$\widehat{D}_{\delta,\eta} = \max_{g\in\mathcal{G}} \sup_{z:\,|\widehat{F}(z)-(1-\delta)|\le\eta} \left|\widehat{F}_g(z) - \widehat{F}(z)\right|, \qquad \eta = 0.04749.$$

A smaller value indicates that bin-specific score distributions align more closely with the overall distribution near the target quantile. We include all 14 bins defined by intersection status, speed, predicted turn, neighboring-agent count and mode entropy.

Local SNSA has a smaller discrepancy at every target (Table 16), indicating more consistent score distributions across the evaluated scene types near the relevant quantiles. These results isolate the unpriced construction; they do not describe the priced scores used in the main table. We separately measure worst-bin coverage error, $\max_g |\widehat{p}_g - (1-\delta)|$, where $\widehat{p}_g$ is the empirical whole-path coverage in bin g at the calibrated threshold. This measures departure from nominal coverage, whereas $\widehat{D}_{\delta,\eta}$ measures differences between score distributions.

Target	Global SNSA	Local SNSA (FacCP)
90%	12.40	**4.73**
95%	7.15	**3.59**
97%	7.04	**2.55**
99%	5.63	**2.08**

Table 16: Quantile-local score discrepancy $\widehat{D}_{\delta,\eta}$ in percentage points. Therefore, lower is better.

Coverage across scene bins. Table 17 reports coverage in all 14 test bins at the 95% target. Local improves intersection coverage, although six-mode coverage for fast-moving agents is slightly lower.

		Single mode		Six modes	
Scene bin	Scenes	Global	Local (FacCP)	Global	Local (FacCP)
Non-intersection	2468	96.83	96.76	95.86	95.05
Intersection	979	93.05	94.77	93.51	94.41
Speed ≤ 5 m/s	1438	95.56	95.13	95.83	95.12
Speed $(5, 12]$ m/s	1513	95.93	97.01	94.63	94.69
Speed > 12 m/s	496	95.78	96.79	95.06	94.68
Turn $\leq 15^\circ$	2843	96.93	96.64	96.00	95.16
Turn $(15, 45]^\circ$	272	94.32	94.65	93.18	93.89
Turn $> 45^\circ$	332	86.90	93.66	89.93	93.15
Neighbors 0–2	1457	93.80	94.97	93.40	93.49
Neighbors 3–6	1279	96.54	96.95	96.14	95.67
Neighbors ≥ 7	711	98.36	97.35	97.15	96.25
Entropy low	1184	97.14	96.60	96.58	96.13
Entropy middle	1074	95.65	96.23	94.73	94.05
Entropy high	1189	94.46	95.75	94.23	94.35

Table 17: Whole-path coverage (%) at the 95% target on internal test scenes, averaged over 16 paired reference-bank draws.

D.10 SINGLE-MODE COMPARISON

Table 19 uses only the top-probability anchor for both reference errors and predictions. It compares the tube construction with joint-trajectory NLE/MVCS baselines without an advantage from extra modes.

Why is the raw-head MTR footprint smaller? All three single-mode methods use the same highest-probability predicted path, but construct different uncertainty regions around it. Raw-head MTR uses its exported covariance and a Gaussian waypoint radius, without calibrating coverage of the complete trajectory. Global and Local CP instead estimate covariance and timestep scales from observed reference errors and calibrate a threshold for simultaneous coverage of all 60 steps. Their larger footprints therefore account for errors that the narrow raw-head ellipses often exclude. At the 95% target, raw-head MTR has a footprint of 26.4 m^2 s and only 1.68% path coverage, compared with 2,362.7/1,882.4 m^2 s and 95.76/96.19% for Global/Local CP. The smaller raw-head footprint is thus accompanied by severe undercoverage. This history also omits the covariance clipping used in training, producing very narrow early ellipses. Here, SD describes variation across scenes, not uncertainty in the reported mean or training instability. A few scenes with very large regions can therefore make SD comparable to, or larger than, the mean. A car moving slowly along a straight lane may have a predictable future, so a small tube can cover its possible positions. At an intersection or during a sharp turn, small differences in speed or turning time can lead to widely separated future positions, requiring a larger region. In six-mode predictions, similar candidate paths overlap heavily and occupy little extra area, while paths branching in different directions occupy more. These differences can produce a wide spread of footprints across scenes, although the exact size also depends on each method's construction. Tables 18– 19 average 16 paired reference-bank draws.

Target	Method	γ	Footprint ± SD	Path (%)
90	Global SNSA	0.00	493.2 ± 113.0	89.86
90	Global SNSA	0.10	419.9 ± 103.1	89.51
90	Local SNSA (FacCP)	0.00	463.4 ± 173.0	89.51
90	Local SNSA (FacCP)	0.10	**386.5** ± 173.9	89.27
95	Global SNSA	0.00	749.5 ± 177.1	95.06
95	Global SNSA	0.10	683.8 ± 143.9	95.19
95	Local SNSA (FacCP)	0.00	680.1 ± 254.9	94.71
95	Local SNSA (FacCP)	0.10	**586.9** ± 243.6	94.87
97	Global SNSA	0.00	985.5 ± 234.7	96.98
97	Global SNSA	0.10	880.0 ± 171.1	97.02
97	Local SNSA (FacCP)	0.00	854.5 ± 320.1	96.68
97	Local SNSA (FacCP)	0.10	**766.1** ± 300.1	96.92
99	Global SNSA	0.00	1,795.5 ± 414.2	99.19
99	Global SNSA	0.10	1,651.1 ± 286.6	99.21
99	Local SNSA (FacCP)	0.00	1,430.5 ± 529.1	99.02
99	Local SNSA (FacCP)	0.10	**1,196.2** ± 432.1	98.85

Table 18: Six-mode pricing comparison. Footprint is in m^2 s. SD is calculated across test scenes.

Target	Method	Footprint ± SD	Path (%)
90	MTR raw-head	20.3 ± 85.4	1.19
90	Global SNSA	1,277.2 ± 0.0	89.98
90	Local SNSA (FacCP)	**1,063.3** ± 330.1	91.30
90	Messoudi NLE	27,369.2 ± 21,075.3	90.71
90	MVCS-L2	84,668.7 ± 108,278.5	91.05
95	MTR raw-head	26.4 ± 111.1	1.68
95	Global SNSA	2,362.7 ± 0.0	95.76
95	Local SNSA (FacCP)	**1,882.4** ± 584.2	96.19
95	Messoudi NLE	47,625.8 ± 36,668.8	95.21
95	MVCS-L2	168,608.7 ± 216,389.7	96.39
97	MTR raw-head	30.9 ± 130.1	2.15
97	Global SNSA	3,356.1 ± 0.0	97.63
97	Local SNSA (FacCP)	**2,472.7** ± 767.6	97.57
97	Messoudi NLE	81,456.4 ± 62,721.4	97.78
97	MVCS-L2	231,938.9 ± 297,170.7	98.08
99	MTR raw-head	40.6 ± 170.8	3.25
99	Global SNSA	5,952.4 ± 0.0	99.06
99	Local SNSA (FacCP)	**3,904.1** ± 1,211.8	98.83
99	Messoudi NLE	150,365.8 ± 115,781.8	99.63
99	MVCS-L2	368,926.5 ± 481,905.0	99.40

Table 19: Single-mode comparison without pricing. Footprint is in m^2 s. SD is calculated across test scenes. NLE and MVCS evaluate membership in joint trajectory ellipsoids.

D.11 WHY JOINT ELLIPSOIDS HAVE LARGE FOOTPRINTS?

Figure 7 illustrates why spatial footprint and joint trajectory volume can rank methods differently. Footprint measures the area available at each timestep and sums these areas over time; it does not capture dependencies between positions at different timesteps. Joint volume instead measures the set of complete trajectories allowed by those dependencies.

For Global and Local SNSA, the single-mode set is a product of 60 timestep ellipses: any sequence of positions lying inside the respective ellipses belongs to the set. NLE and MVCS instead use a joint ellipsoid that couples all 120 trajectory coordinates. Its projection at a given timestep can be large, even though positions chosen independently from different projections may not form an admissible trajectory. A thin, tilted ellipse provides a simple analogy: it can have wide projections onto both axes while enclosing little area. Cross-time dependence produces the same effect in the joint trajectory ellipsoid.

In Figure 7, all four methods cover the observed trajectory. Global and Local footprints are 3,338.0 and 1,703.0 m^2 s, compared with 29,011.9 and 52,215.6 for NLE and MVCS. Their joint log-volumes, $\log(V_{120}/\mathrm{m}^{120})$, are respectively 297.61, 255.98, -182.06 and -117.25. Thus, NLE and MVCS allow a wider spatial region at individual timesteps but a smaller volume of complete trajectories. Neither metric determines the other: footprint describes spatial occupancy over time, whereas joint volume also reflects restrictions on how positions can vary together.

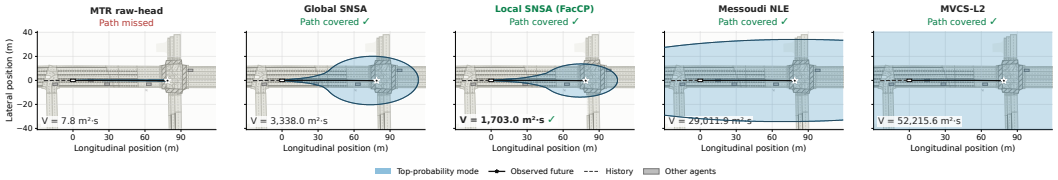


Figure 7: Single-mode example at the 97% target. FacCP covers the whole path with 49.0% less footprint than Global SNSA. NLE and MVCS show spatial projections of joint trajectory ellipsoids.

Table 20: Comparison with OT-CP+ on planar quadcopter trajectories. Coverage denotes simultaneous trajectory coverage. Full log-volume is the mean per-timestep log-volume of the complete prediction region. Clipped log-volume is estimated within a common finite reference-bank bounding box and is not the full region volume. Calibration denotes total calibration time (Cal. (s)). Membership and geometry times (Member. (ms) and Geom. (ms) respectively) are measured per query. Lower volume and runtime are better.

Method	Cov. (%)	$\log V$	Clipped $\log V$	Cal. (s)	Member. (ms)	Geom. (ms)
FacCP	**95.31**	**-32.309**	**-32.342**	**0.32**	**0.019**	**1.24**
OT-CP+ full trajectory	100.00	∞	-21.812	1.12	0.082	6.28
OT-CP+ per timestep	92.97	∞	-21.835	94.04	3.733	374.83
FacCP-weighted OT	97.66	∞	-21.811	153.26	3.773	611.85

D.12 What changes when pricing is added?

Pricing uses the forecaster's candidate probabilities to allocate tube radii. We add $b_k = -\gamma \log \pi_k$ to each candidate's score, so a high-probability candidate incurs a smaller penalty and retains more of the calibrated radius budget. A low-probability candidate incurs a larger penalty and receives a smaller budget, potentially becoming inactive. This favors candidates that the forecaster regards as more plausible, reducing the footprint spent on unlikely alternatives. The allocation changes uncertainty around the predicted paths; it does not move the paths themselves.

We calibrate the resulting scene score on a separate calibration set, preserving the marginal whole-path coverage guarantee under exchangeability. This guarantee concerns coverage by at least one candidate tube across scenes; it does not require equal protection for every candidate or scene group. Pricing therefore offers a probability-informed allocation, rather than a guarantee of balanced coverage across high- and low-probability outcomes.

Table 18 isolates this effect by holding the encoder, covariance matrices and SNSA scales fixed and comparing $\gamma = 0$ with $\gamma = .10$. At the 95% target, pricing reduces Local's mean footprint from 680.1 to 586.9 m^2 s, while empirical whole-path coverage changes from 94.71% to 94.87%.

E Additional Baselines

Planar Quadcopter: OT-CP+ (Thurin et al., 2025). We compare FacCP against OT-CP+ on the planar quadcopter task. We consider full-trajectory OT-CP+, which constructs a single transport rank map over the complete residual trajectory, and a per-timestep adaptation, which fits separate low-dimensional transport maps and combines their scores using simultaneous trajectory-level conformal calibration. We additionally evaluate OT with FacCP's learned energy-based retrieval weights and with uniform weights over nearest neighbors in the learned representation. All methods target 95% simultaneous trajectory coverage.

Finite-reference limitations of trajectory-level OT: Full-trajectory OT-CP+ treats the complete residual trajectory as one high-dimensional vector and estimates a single transport rank map. In the finite-reference implementation, the rank is determined by the maximizer among k affine functions. Their pairwise differences span at most $k - 1$ directions, leaving a nullspace of dimension at least $d_R - k + 1$ when $k \leq d_R$. The rank is constant along this nullspace, and every nonempty prediction region therefore extends indefinitely in those directions. This is a structural limitation of the finite-reference rank construction rather than a numerical optimization failure. Consequently, full-trajectory OT-CP+ achieves 100% empirical coverage but has infinite full log-volume. Even after clipping the regions to a common bounded evaluation domain, its estimated log-volume is -21.812, compared with -32.342 for FacCP. This illustrates how finite-reference transport ranks can produce highly conservative prediction regions in high-dimensional trajectory spaces.

Per-timestep OT and computational scalability: Per-timestep OT-CP+ provides an intermediate approach between full-trajectory transport and FacCP. Instead of fitting one high-dimensional transport map, it fits a separate 6-dimensional OT map at each timestep. The resulting scores are combined through a maximum-over-time statistic and calibrated using a single trajectory-level conformal

Table 21: Comparison with CVQR on planar quadcopter trajectories, targeting 95% simultaneous coverage. $\log V$ denotes full-trajectory log-volume normalized by horizon length. Calibration is total calibration time; membership and geometry times are per query. Lower volume and runtime are better.

Method	Cov. (%)	$\log V$	Cal. (s)	Member. (ms)	Geom. (ms)
FacCP	95.70	-34.899	**3.75**	**0.112**	**1.77**
CVQR	96.88	$\mathbf{-34.948}$	10.43	42.77	269.02

threshold. This avoids the specific rank deficiency of full-trajectory OT but requires repeated transport optimization for every query. Its finite-reference Laguerre cells can also remain unbounded.

The computational cost is substantial. Per-timestep OT-CP+ requires 374.83 ms/query and 94.04 s for calibration, compared with 1.24 ms/query and 0.32 s for FacCP, while achieving 92.97% versus 95.31% coverage. Increasing the reference support further increases the cost of the OT problems.

FacCP instead learns an uncertainty-relevant retrieval representation using the energy score. This representation identifies reference trajectories with similar conditional residual laws without relying on proximity in the original high-dimensional query space. The retrieved weights are shared across timesteps, and the corresponding local means and covariances are computed efficiently using GPU-accelerated weighted operations. Regularized per-timestep ellipsoids then provide compact, bounded prediction regions without solving query-specific transport problems.

Applying FacCP's energy-based retrieval weights to OT does not eliminate the limitations of the finite-reference rank construction. The weighted OT variant retains infinite full log-volume and requires 611.85 ms/query, despite achieving 97.66% empirical coverage. Thus, improving residual-law localization alone does not resolve the computational and geometric bottlenecks of this OT implementation.

These results reinforce that learning locally appropriate geometry and calibrating globally is useful, but the choice of geometry constructor still matters enormously in high dimensions.

Planar Quadcopter: CVQR (Kondratyev et al., 2026). We compare FacCP against full-trajectory Conformal Vector Quantile Regression (CVQR) on the planar quadcopter task , targeting 95% simultaneous trajectory coverage. CVQR learns conditional residual geometry through a conditional convex potential, while FacCP constructs explicit local ellipsoids from energy-weighted residual moments. Both methods use split-conformal calibration.

Numerical validation and volume estimation: CVQR uses double-precision inversion of its learned conditional convex map, with amortized warm starts, L-BFGS optimization, and Newton refinement. All accepted inversions satisfy residual tolerances below 10^{-5}. Full-trajectory volumes are estimated using Jacobian Monte Carlo integration, with relative standard errors of 3.6–6.7%.

Computational comparison: CVQR achieves 96.88% coverage and -34.948 normalized log-volume, compared with 95.70% and -34.899 for FacCP. However, CVQR requires iterative inverse-map evaluation for membership, taking 42.77 ms/query versus 0.112 ms/query for FacCP ($382\times$ faster). Constructing CVQR geometry from 256 numerically verified boundary samples takes 269.02 ms/query, compared with 1.77 ms/query for GPU-computed FacCP ellipsoids ($152\times$ faster). Calibration takes 10.43 s versus 3.75 s.

FacCP additionally uses approximately half as many parameters (101k versus 201k). Its learned retrieval weights directly determine local residual moments, enabling explicit, bounded ellipsoidal geometry without iterative inversion or boundary sampling. These results demonstrate substantial computational advantages while maintaining comparable empirical coverage and normalized log-volume.

Planar Quadcopter: GRCP (Conrad et al., 2026). We compare FacCP against Geometric Rank Conformal Prediction (GRCP) and a FacCP-weighted GRCP variant on planar quadcopter trajectories ($H = 100$), targeting 95% simultaneous trajectory coverage.

Table 22: Comparison with GRCP on planar quadcopter trajectories ($H = 100$), targeting 95% simultaneous coverage. $\log V$ denotes full-trajectory log-volume normalized by horizon length. Calibration denotes total calibration time; geometry representation and membership times are per query. Lower volume and runtime are better.

Method	Cov. (%)	$\log V$	Cal. (s)	Geom. (ms)	Member. (ms)
FacCP ellipsoids	95.31	$\mathbf{-32.309}$	0.329	1.240	**0.019**
GRCP	96.88	-28.254	**0.039**	**0.075**	0.091
FacCP-weighted GRCP	96.88	-29.854	0.158	0.517	0.110

GRCP constructs nonconformity scores from spatial-rank vector norms. We adapt it to trajectories by computing spatial ranks separately for each six-dimensional timestep residual, after whitening using statistics estimated from the geometry bank. Standard GRCP uses uniform weights over 256 nearest neighbors selected from standardized nominal states and controls. FacCP-weighted GRCP instead selects neighbors in the learned uncertainty representation and uses normalized energy-based retrieval weights. Both variants use timestep normalization scales estimated from a separate subset, followed by maximum-over-time scoring and a single global split-conformal calibration threshold.

Full-volume estimation: GRCP prediction regions are defined implicitly by spatial-rank membership tests. We estimate their full volumes using importance sampling with two independent, full-support Student-t proposals. For each query and timestep, proposal distributions are fitted using approximate boundary points, with 16,384 samples drawn from each proposal. Membership indicators divided by proposal densities provide volume estimates, transformed from whitened to physical coordinates. Trajectory log-volume is obtained by summing timestep log-volumes. The two proposals agree, and maximum estimated timestep relative errors are 1.29% for GRCP and 1.61% for FacCP-weighted GRCP.

Geometry and retrieval comparison: Standard GRCP achieves 96.88% coverage with normalized log-volume -28.254. Replacing its input-space localization with FacCP's learned energy-based retrieval improves this to -29.854 at identical empirical coverage. This improvement demonstrates that the learned representation benefits spatial-rank geometry, independently of the ellipsoidal constructor used by FacCP.

FacCP ellipsoids achieve 95.31% coverage and a substantially smaller normalized log-volume of -32.309. Although GRCP has lower calibration and geometry-representation times, FacCP provides faster membership evaluation (0.019 versus 0.091–0.110 ms/query) through explicit ellipsoidal geometry. Thus, learned residual-law retrieval improves alternative geometry constructors, while the choice of constructor remains important for prediction-set efficiency and computational cost.

Real Datasets: NYC Taxi. We evaluate FacCP on the NYC taxi dataset Dheur et al. (2025b), where four trip-level features, including pickup location, are used to predict the two-dimensional drop-off coordinates. The destination distribution is spatially heterogeneous and potentially multimodal, making non-ellipsoidal prediction regions particularly useful. Table 23 and Fig. 8 compare five methods at a nominal 95% coverage level. FacCP Gaussian Splats achieves 95.33% coverage with an average area of 126.39 km^2, reducing area by 22.8% relative to covariance prediction and 41.8% relative to resCONTRA. FacCP weighted KDE also attains 95.83% coverage with an area of 141.23 km^2. Both FacCP variants require only 4,612 learned parameters, versus 266,757 for covariance prediction and 16.9 million for resCONTRA. Although the uncalibrated KDE baseline produces smaller regions, its empirical coverage is only 93.75%. Importantly, weighted KDE and Gaussian Splats define nested sets through level thresholds on their query-dependent scores. They therefore admit the same global split-conformal calibration as ellipsoids, preserving marginal validity while allowing flexible, potentially disconnected prediction regions without retraining the retrieval representation.

Real Datasets: River Flow. We further evaluate FacCP on the river-flow prediction benchmark Dheur et al. (2025b), where historical streamflow measurements are used to forecast future flows at monitoring sites in the Mississippi River network. The task involves correlated outputs whose uncertainty can vary with hydrological conditions; we consider first four targets in the source ordering for set prediction. Table 24 reports results at a nominal 95% coverage level. FacCP ellipsoids

Table 23: NYC taxi drop-off prediction at nominal 95% coverage. Area is the mean prediction-region area in km^2. KDE (95% mass) is uncalibrated. Lower area and parameter count are better at comparable coverage.

Method	Coverage (%)	Area (km^2)	Parameters
KDE (95% mass)	93.75	**93.20**	0
Covariance prediction	94.00	163.73	266,757
resCONTRA	94.00	217.16	16,875,536
FacCP weighted KDE	**95.83**	141.23	**4,612**
FacCP Gaussian Splats	95.33	**126.39**	**4,612**

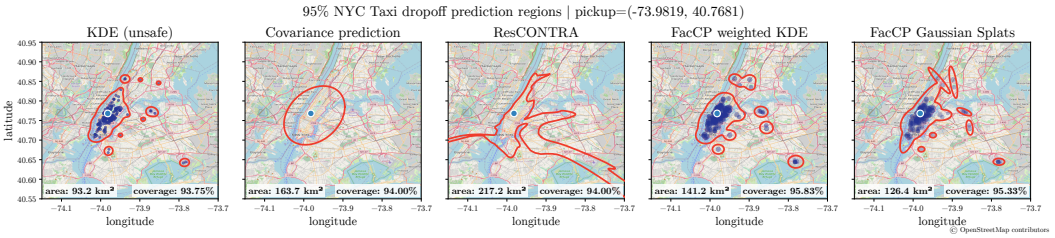


Figure 8: NYC taxi dropoff prediction regions at a nominal 95% level. Blue marks the pickup; red outlines the regions. Insets report test coverage and region area.

achieve 95.89% coverage with a mean 4-D volume of 78.52, compared with 137.63 for resCONTRA and 172.94 for covariance prediction. This corresponds to volume reductions of 42.9% and 54.6%, respectively, at comparable coverage. FacCP also uses only 21,264 learned parameters, compared with 4.4 million for resCONTRA and 280,590 for covariance prediction. These results demonstrate that learned local residual geometry can improve both set efficiency and parameter efficiency on real-world multivariate regression.

Table 24: River-flow prediction at nominal 95% coverage. Mean 4-D prediction-region volume and learned parameter count are reported. Lower volume and parameter count are better at comparable coverage.

Method	Coverage (%)	Mean volume	Parameters
FacCP	**95.89**	**78.52**	**21,264**
resCONTRA	94.95	137.63	4,407,328
Covariance prediction	95.72	172.94	280,590